STOL Aircraft Design Project

Jacob Giancola, Varun Dhond, Matthew Breitstone, Zane Gilmore
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This report outlines the entire design process of our aircraft, including the decisions and techniques that we employed to reach our final design. The problem that we will be addressing is the need for quick access to remote communities north of the Arctic Circle. These small communities typically rely on aircraft for transportation, transportation of goods, and medical transport. Our objective is to design small, affordable aircraft that can serve these small communities. Using existing aircraft, such as the Cessna Caravan, we generated initial assumptions including, but not limited to, a turboprop engine, a fixed landing gear, and an upper wing design. We continued to generate concepts that used opposing features to our initial assumptions to find the best design solution. Using MATLAB modeling and external resources, we were able to compute design parameters and estimate the performance of our aircraft. Using our design goal of 2,500 pounds, we were able to compute our gross takeoff mass to be 5,861 pounds. Building off this data, we found that our estimated takeoff distance is 658 feet and our landing distance is 769 feet. Using the Pratt & Whitney PT6A-140 turboprop engine, our aircraft will have 867 SHP or 647 kW. The use of this engine will allow our aircraft to fly at a maximum speed of Mach 0.3 and reach up to 850 nautical miles in range over an endurance time of 10 hours. This speed and power allow our aircraft to successfully clear the minimum rate of climb required by the external flight parameters. Our aircraft will also feature state-of-the-art avionics to allow for complex navigation through harsh Arctic conditions. Advanced GPS and landing instruments such as an ILS receiver will be necessary for landing in low visibility conditions. These measurable results indicate a sound design concept for a short takeoff and landing aircraft that will be FAA 14 CFR Part 23 certifiable.

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1 INTRODUCTION

Project Report Introduction

The overall scope of this report is to outline the technical steps that we took to design our short takeoff and landing (STOL) aircraft. Throughout this report you will be able to read the step-by-step process that our team took to design our aircraft. The problem will be defined as will the steps we took to solve this problem. Each technical decision will also be described in detail with accompanying tables and graphs to justify these decisions. Each decision was meticulously made to adapt the aircraft to the problems that we faced and to make the aircraft perform at the highest possible level to ensure mission success.

Problem Statement

Communities north of the Arctic Circle rely heavily on aircraft for transportation, cargo delivery, and emergency medical services due to their remote locations and limited ground infrastructure. The harsh Arctic environment, including extreme cold temperatures, icing conditions, poor visibility, and unprepared runways, makes reliable and affordable aircraft operations difficult. The problem to be solved is the need for cost-effective, reliable, and versatile aircraft capable of short takeoff and landing on snow or ice-covered fields while safely operating in icing and low visibility conditions. The aircraft must support passenger, cargo, and medical missions with minimal reconfiguration and limited ground support to improve year-round accessibility and safety for Arctic communities.

Problem Scope

The scope of the problem we are trying to address is the conceptual design of a fixed-wing aircraft capable of operating north of the arctic circle with short takeoff and landing capability. The design focuses on meeting passenger, cargo, and aerial ambulance mission requirements while complying with FAA 14 CFR Part 23 certification standards. While we are primarily focusing on the conceptual design of this aircraft, the cost analysis, structural design, and manufacturing are outside of the scope of this report.

Initial Assumptions

Our initial research started with analysis of existing aircraft with similar capabilities to what our aircraft will possess. To gather our initial assumptions, we researched the DHC-3 Otter, Pilatus PC-6 Porter, and the Cessna 208 Caravan. From this, we gathered that it would be efficient to feature a turboprop engine in a high wing and fixed landing gear configuration. These reference aircraft demonstrate the effectiveness of these design features in rugged terrain and harsh weather conditions. To fulfill the mission requirements, we wanted to focus on the loading and unloading of the aircraft and how to make it as easy as possible for ground crew. Using these initial design features will help us grow our aircraft and learn what is best for flight performance and mission performance.



Figure 1. DHC-3 Otter Aircraft [DE HAVILLAND DHC-3 OTTER]

For the flight performance of our aircraft, we would like to target an 800 nautical mile range to allow for flexibility in the routing of our aircraft. A larger range unlocks more of the Arctic communities that we can support, and the more bases we can supply these communities from outside of the Arctic Circle. We also targeted a cruising speed of Mach 0.4 to ensure smooth flight and timely delivery of cargo. Faster transport could also potentially save the life of someone while the aircraft is undergoing medical transport. The engine selection will also play into these considerations and assumptions later in this report.

Our aircraft, given these initial assumptions, will be compliant with FAA 14 CFR Part 23 certification standards. The decisions made after assuming these initial target parameters will be made with Part 23 standards in mind to allow for the safe and compliant design of our aircraft.

Mission Statement

We are to design and deliver reliable, fixed-wing Short Takeoff and Landing (STOL) aircraft capable of safely transporting passengers, cargo, and emergency medical payloads to and from remote communities north of the Arctic Circle. The system will operate year-round on short, snowy, ice-covered, and unprepared fields, while withstanding extreme cold and known icing conditions. The aircraft will provide dependable access with minimal infrastructure support while minimizing the overall cost of transporting people, goods, and medical services to isolated regions.

For this example, our aircraft will be flying a cargo transport mission. The segments of the mission will occur within a range of 850 nmi at a cruise altitude of approximately 20,000 ft ASL. Our aircraft will be able to carry an 8,000-watt generator along with 55 gallons of diesel fuel in a 9-passenger configuration. We will estimate the total payload to be 2,500 lbs. This mission will consist of the following segments: takeoff, climb, cruise, loiter, and landing phase. These will be the mission segments for the inbound and outbound legs of the mission. The mission will be supplied with an 8% fuel reserve and enough fuel to sustain a 1-hour endurance of the loiter phase (30 minutes per loiter phase). Below is a simple mission diagram depicting the phases of the mission at hand.

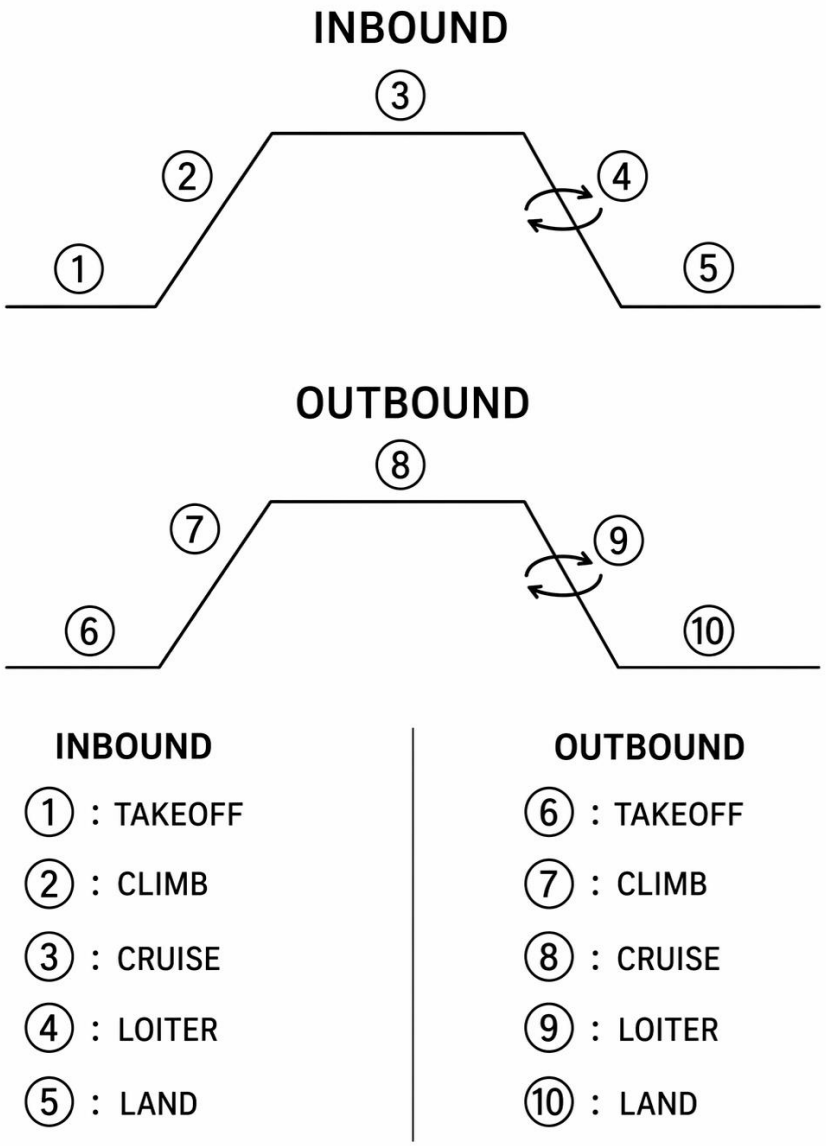


Figure 2. *Simple Mission Phase Diagram [Digitized with ChatGPT]*

2 NEEDS, REQUIREMENTS, AND RISK ANALYSIS

The design of this project must be designed to fit the needs, requirements, and risk analysis presented by the stakeholders and Federal Aviation Administration safety guidelines. The stakeholders have presented their needs for this aircraft that must be considered throughout this design process. The requirements of this project have been designated as well to safely match the mission design. To fulfill the mission profile, aircraft risks and mitigation must be accounted for to ensure safe operating conditions and design parameters. The aircraft

should account for risks that can occur due to weather conditions, altitude conditions, and speed conditions. To ensure that the aircraft will operate nominally, these risks must be evaluated before and after each design of the aircraft to verify its safety and suitable mission goals. The major risks with this specific mission were concerns due to the low temperatures that this aircraft will be operating on.

2.1 STAKEHOLDERS AND THEIR NEEDS

Our stakeholders were identified through a mission-driven and systems approach starting from the aircraft's intended environment and expanding outward. First, we analyzed the mission requirements that were outlined in the project description. We thought of who would be directly impacted and dependent on this aircraft to accomplish this mission. This led us to identify the primary stakeholders for this mission. We identified our primary stakeholders as:

- Remote Community Residents
    - Rely on the aircraft for safe, reliable, and affordable transportation between isolated communities.
- Cargo and Supply Operators
    - Use the aircraft to deliver essential goods and equipment to areas with little infrastructure.
- Pilots
    - Operate the aircraft in extreme conditions, so handling, visibility, and avionics directly affect safety.
- Aircraft Operators
    - Own and manage the aircraft; operating cost, reliability, and mission flexibility determine profitability.

For our secondary stakeholders, we considered ownership and operational responsibility through examining who is responsible for operating the aircraft. We also thought of who is responsible for the maintenance of the aircraft over its lifecycle and who ensures the aircraft remains operational. We identified the secondary stakeholders as:

- Aircraft mechanics
    - Maintain and repair the aircraft in harsh environments, so durability and accessibility affect uptime.
- Federal Aviation Administration (FAA)
    - Set certification and safety standards that the aircraft must meet legally operate.

We identified each stakeholder's need through a breakdown of their requirements based on the mission profile and operational environments. First, we started from the mission definition that requires the aircraft to be passenger transport, cargo delivery, and aerial ambulance. We analyzed each stakeholder, figuring out their needs from the aircraft to fulfill their role in the mission. We then used the operating environment, such as the temperature, the runway infrastructure, and remoteness, to add to each stakeholder's needs. Our stakeholder needs were identified as:

- Remote Community Residents
    - The aircraft shall provide safe transportation between remote communities.
    - The aircraft shall operate reliably year-round in Arctic weather.
    - The aircraft shall provide affordable travel for residents of remote regions.
    - The aircraft shall allow passengers to travel with personal baggage and essential items.
    - The aircraft shall provide a comfortable and secure cabin environment during flight.
    - The aircraft shall minimize travel disruptions due to weather or runway conditions.
- Cargo and Supply Operators
    - The aircraft shall transport essential supplies and equipment to remote communities.
    - The aircraft shall accommodate heavy, bulky, or irregularly shaped cargo.
    - The aircraft shall allow cargo loading and unloading with minimal ground infrastructure.

    - The aircraft shall operate efficiently while carrying full cargo payloads.
    - The aircraft shall enable quick turnaround between cargo missions.
- Pilots
    - The aircraft shall provide predictable and stable handling, especially at low speeds.
    - The aircraft shall support safe operations in low visibility and extreme lighting conditions.
    - The aircraft shall reduce pilot workload during takeoff and landing.
    - The aircraft shall allow safe operation in icing and severe cold weather.
    - The aircraft shall provide adequate situational awareness during ground and flight operations.
- Aircraft Operators
    - The aircraft shall minimize operating costs per mission.
    - The aircraft shall maintain high dispatch reliability in Arctic environments.
    - The aircraft shall support multiple mission types with minimal reconfiguration time.
    - The aircraft shall operate from short, snow- or ice-covered runways.
    - The aircraft shall provide sufficient range to complete round-trip missions without refueling.
- Maintenance Technicians
    - The aircraft shall allow maintenance in remote areas with limited facilities.
    - The aircraft shall provide easy access to critical systems and components.
    - The aircraft shall use systems that are durable in cold, wet, and corrosive environments.
    - The aircraft shall minimize the time required for inspection and routine maintenance.
    - The aircraft shall minimize unscheduled maintenance events.
- Federal Aviation Administration (FAA)
    - The aircraft shall comply with FAA 14 CFR Part 23 airworthiness standards.
    - The aircraft shall demonstrate safe operation in known icing conditions.

- The aircraft shall meet structural and system safety requirements.
- The aircraft shall include approved communication and navigation capabilities.
- The aircraft shall meet operational safety standards for flights in remote regions.

2.2 REQUIREMENTS

1. Our aircraft shall accommodate a minimum of 2 pilots to ensure safe operations and to comply with mission requirements specified in the project design brief. This requirement is stated by the project specifications and allows sufficient workload distribution for each pilot in the challenging arctic conditions, including low visibility and icing environments. This can be verified by confirming that the aircraft design includes seating and controls for 2 pilots, a flight deck layout supporting dual operation, and accommodating the weight for both crew members. This is not dependent on the aircraft design variables, as it is a fixed constraint specified by the project requirements.
2. Our aircraft shall accommodate 9 passengers, taking luggage into account, and a minimum seat pitch of 31". This is verifiable by checking cabin layout dimensions and confirming that the payload capacity can support 9 passengers and their luggage. This requirement is dependent on the maximum payload weight, aircraft design limits, and fuel fraction. The passenger capacity and cabin layout were selected based on comparable aircraft such as the Cessna 208B Grand Caravan EX (Textron Aviation, n.d.) and Pilatus PC-6 Porter (Pilatus Aircraft Ltd., n.d.), which have similar mission profiles. A 31" seat pitch was chosen within the standard industry range (28-34") to balance comfort and capacity (Boeing, n.d.), while payload assumptions were based on FAA standard passenger and baggage weights (Federal Aviation Administration, 2012).
3. Our aircraft shall support a minimum payload capacity of 2,500 lbs (estimated from standard passenger, fuel, and equipment weights), representing cargo equivalent to 9 passengers with baggage, an 8,000-watt generator, and 55 gallons of diesel fuel, to

allow cargo transport to remote regions in the Arctic. The estimated value of the 9 passengers is 200 lbs per passenger, making the total 1800 lbs. The diesel fuel is estimated to be 7 lbs/gal, making 55 gallons of fuel equal to 385 lbs total. The generator is estimated to be 200 lbs. Adding these estimations we get 2,385 lbs, rounding to 2,500 lbs shows the estimation level of precision we have obtained. This is verifiable by calculating the maximum payload capacity from the weight breakdown:

$$W_{payload,\,max} = W_0 - W_e - W_f$$

And confirming that:

$$W_{payload,\,max} \geq W_{required\ payload}$$

Where the required payload includes cargo equivalent to 9 passengers with baggage, an 8,000-watt generator, and 55 gallons of diesel fuel. This requirement depends on aircraft sizing variables including the maximum takeoff weight, empty weight fraction, fuel fraction, mission range, wing loading, lift capacity, and engine power constraints.

4. Our aircraft shall support a minimum payload capacity of 1,300 lbs. This is represented by 4 passengers, one 150 lb medical litter, and 300 lbs of medical supplies, all to allow aerial ambulance operations in remote arctic regions. The payload minimum is estimated to use the same estimation per passenger as requirement 3, totaling 800 lbs. Combined with a 150 lb litter and 300 lbs of medical supplies, that brings the estimate to 1,250 lbs. We rounded up due to the level of precision we have due to the estimates of all sections of this mission. This requirement is verifiable by calculating the maximum payload capacity with the following equation,

$$W_{payload,\,max} = W_0 - W_e - W_f$$

While confirming that,

$$W_{payload,\,max} \geq 1{,}300\ lbs$$

This requirement depends on some aircraft design parameters including maximum takeoff weight, empty weight fraction, fuel weight, mission range, cabin configuration constraints, aerodynamic performance, and engine power limitations.

5. Our aircraft shall achieve a maximum takeoff distance over a 50 ft obstacle of 800 ft or less and a maximum landing distance over a 50 ft obstacle of no more than 800 ft to ensure safe operation from short and obstructed runways in the remote regions. This requirement is verifiable by calculating takeoff and landing performance using equations like thrust to weight ratio, wing loading, and lift coefficients (Raymer, 2018). The 50 ft obstacle clearance metric and target distances are based on standard aircraft performance definitions and informed by comparable STOL aircraft such as the Cessna 208B Grand Caravan EX and Pilatus PC-6 Porter (Textron Aviation, n.d.; Pilatus Aircraft Ltd., n.d.).
6. Our aircraft shall be capable of operating safely in the arctic environment, including a minimum operating temperature of −40°C, operation on snow- or ice-covered runways, and functionality under reduced visibility conditions to allow mission capability in remote regions. These conditions include maintaining structural integrity, system functionality, and controllability in extreme cold and rough surface conditions. A minimum operating temperature of −40°C and a runway friction coefficient of approx. 0.25 are assumed to represent arctic conditions, consistent with typical cold weather aviation environments. This is verifiable by evaluating the system's functions and performance at low temperatures, as well as assessing takeoff, landing, and braking performance on low friction surfaces for safety. This requirement depends on design parameters such as material properties at low temperatures, engine cold-start capability, de-icing and anti-icing systems, tire-runway friction characteristics, aerodynamic performance under icing conditions, and system reliability.
7. Our aircraft shall have a minimum cruise range of 400 nmi for one bound of the flight, with sufficient fuel capacity to complete both inbound and outbound without refueling to make sure there are reliable operations between remote arctic locations. This requirement is based on mission specifications that require transport between isolated communities with limited infrastructure. This requirement can be verified using the Breguet range equation and power specific fuel fraction calculations:

$$R = \frac{V}{c_p}\frac{L}{D}\ln\left(\frac{W_i}{W_f}\right)$$

While confirming that the calculated range is greater than or equal to 800 nmi due to the minimum range of one bound of the flight being 400 nmi, allowing inbound and outbound flights without refuel. This is dependent on many aspects of the aircraft's design and performance like the fuel fraction, total fuel weight, lift to drag ratio, engine specific fuel consumption, cruise velocity, aircraft weight fractions, and aerodynamic efficiency and propulsion and system performance.

8. Our aircraft shall be capable of safely operating in arctic conditions, including flight in known icing conditions, and operating under extreme visibility conditions like no sun winters and continuous daylight summers, to ensure reliable missions in remote regions. This requirement is verifiable by using appropriate de-icing and anti-icing systems, as well as navigation and guidance systems capable of maintaining safe operations in low visibility environments. This requirement depends on the system design parameters including de-icing and anti-icing system capabilities, sensor and avionics performance, power availability, and overall system reliability under harsh arctic conditions.

The hardest requirements that will have to be met are the STOL performance, cargo payload, and range without refueling. This is because they all control aircraft sizing variables like maximum takeoff weight, wing size, and fuel fraction, and they are pulling the design in opposite directions. The requirements that will fight each other are combinations of the hardest requirements to meet due to their opposing needs for success of the missions. Additionally, this is all made significantly worse simultaneously by the arctic conditions.

2.3 PRELIMINARY RISK ANALYSIS

The first risk analyzed was airframe icing leading to loss of lift of the aircraft. This problem is prevalent because during climb, cruise, or descent supercooled droplets through moisture at sub-freezing temperatures may occur on aircraft parts such as wing leading edges, horizontal stabilizer, propeller blades, and engine inlets. The effect of icing is that it could increase the stall speed, drag, required thrust, and control surface hinge movements. This

could result in loss of control causing Controlled Flight into Terrain (CFIT). The most vulnerable phases are during low-altitude approaches into remote Arctic airstrips and holding in icing layers. Our mitigation of this risk is to use certified thermal or pneumatic de-icing system, ice detection sensors, increased power margin, higher wing loading margin, and conservative stall speed sizing.

The second risk we analyzed was an engine cold-soak startup failure. At extreme temperatures less than or equal to -40°C, oil viscosity increases, fuel gelling may occur, battery capacity drops, and metal contraction which can affect tolerances. These can result in inability to start engine, reduced power output, and compressor surges. The most vulnerable phase that this could occur is during ground turnaround at remote strips and takeoff after overnight exposure. This could result in the aircraft being stranded in remote areas and potentially risk passenger survival. Our mitigating strategies are to include engine block heaters, Arctic-rated lubricants, auxiliary power units (APU), utilize redundant battery systems, and cold-start qualification testing.

A third risk that was analyzed was runway overrun on a short unprepared surface. The problem associated with this is that during operations in gravel, snow, or ice strips, there are reduced braking coefficients, limited runway lengths, and crosswind gusts. These can cause landing overruns, nose gear collapse, and loss of directional control. The way to mitigate this issue is to create an STOL wing design, high-lift devices, and oversized tundra tires.

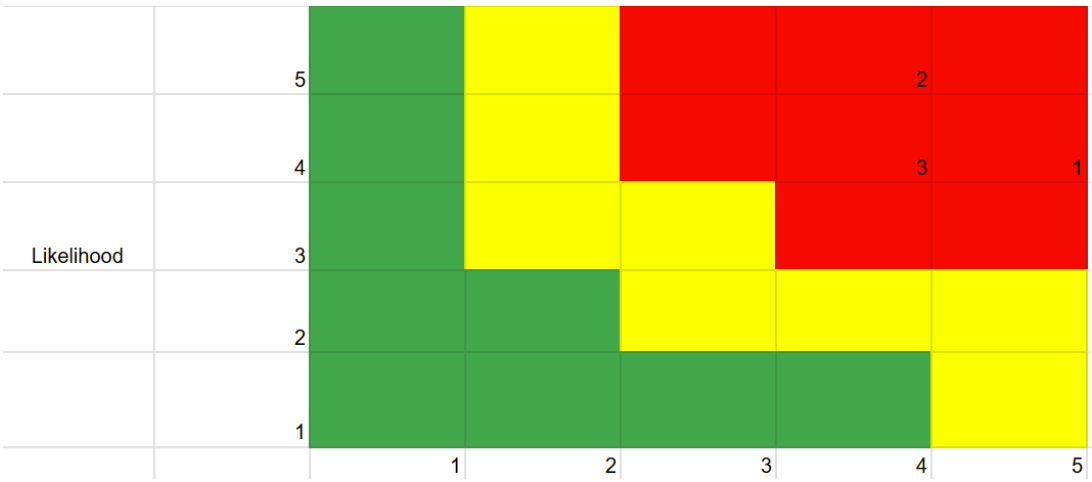


Figure 3. Risk Matrix Before Mitigation

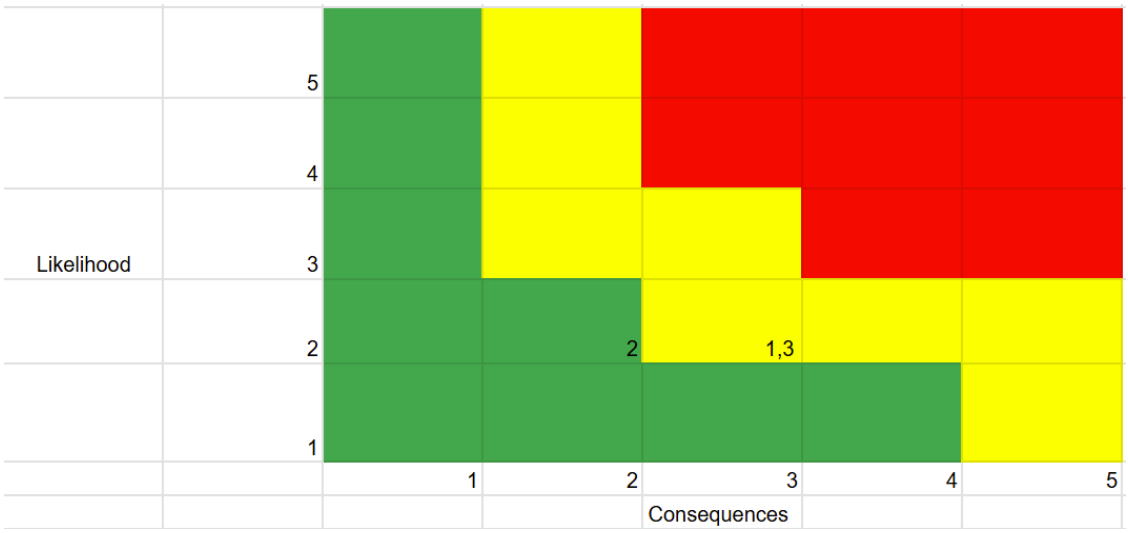


Figure 4. Risk Matrix After Mitigation

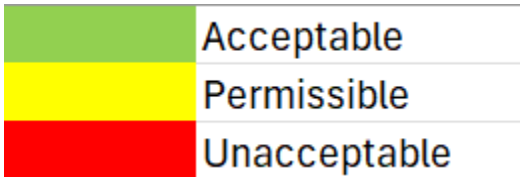


Figure 5. Risk Matrix Legend

3 Estimating Design Parameters

This section will introduce and describe the methods that we use to estimate our design parameters. Many of the design parameters are inspired by existing aircraft that are/can be used for the same purposes. Here, the bulk of our design will be described by these parameters.

3.1 EXISTING DESIGNS

Aircraft 1: DHC-3 Otter



Figure 6. DHC-3 Otter Aircraft [DE HAVILLAND DHC-3 OTTER]

- Versatile wheel, float, and ski configurations
- 58 ft wingspan
- 8,000lb max weight
- 120 kts cruise speed
- Maximum payload weight: 2,100 lb
- Turboprop
- Capacity: 9-11 passengers
- Length: 12.75 m (41 ft 10 in)
- Range: 820 Nautical miles

- Endurance: 9.5 hours

Aircraft 2: Cessna 208B Grand Caravan EX



Figure 7. *Cessna 208B Grand Caravan EX [https://www.bjtonline.com/aircraft/cessna-208b-grand-caravan-ex]*

- Range: 494 Nautical miles
- Endurance: 5 hours
- Turboprop engine
- 194 kts Max cruise speed
- Balanced Field Length: 2,742 ft
- Max payload weight 3,600 lbs
- Maximum Takeoff Weight: 8,807 lb
- Rate of Climb: 1,330 fpm

Aircraft 3: Pilatus PC-6 Porter



Figure 8. Pilatus PC-6 Porter [https://www.flyingbulls.at/en/fleet/pilatus-porter-pc-6]

- 10 passenger capacity
- Range: 870 Nautical miles
- Endurance: 4 hours
- Turboprop
- Height: 10.5 ft
- Width: 51.8 ft
- Length: 35.8 ft
- Takeoff distance: ~640–800 ft

- Landing distance: ~600–800 ft
- Useful load: ~2,500 lb

3.2 ESTIMATING GROSS TAKEOFF WEIGHT

The Arctic mission requirements include short-field takeoff, moderate-range cruise, and reserve capability. The mission consists of takeoff from an unprepared runway, climb to cruise altitude, a 400–450 nmi cruise segment, followed by loiter, descent, and landing, with additional reserve climb and loiter segments to ensure safety in uncertain conditions. These mission phases determine fuel fraction, wing loading, and overall takeoff weight. Our strategy focuses on maximizing mission reliability and versatility rather than speed by prioritizing STOL performance, payload capability, and operational flexibility. This is achieved through a low-wing loading, high-lift airfoil, and efficient high-aspect-ratio wing, combined with a turboprop propulsion system for strong low-speed thrust and fuel efficiency. This approach ensures that the aircraft can consistently complete missions in harsh Arctic environments where reliability and performance margins are more critical than maximum speed.

To size the aircraft, we first estimated the cruise Mach number to be 0.4. We came to this number by referring to Raymer, *Aircraft Design: A Conceptual Approach*, a typical turboprop cruise Mach range of (0.4–0.5), and referencing our similar aircraft, the DHC-6 Twin Otter and the Cessna 208 Caravan. The mission emphasizes reliability and short-field performance over high cruise speed so being comparable to current regional turboprop aircraft operating in remote environments (Twin Otter, Caravan) that cruise between Mach 0.40–0.50, our estimate makes sense.

For our range, we estimated our value with the DHC-6 Twin Otter range capability, the ATR 42 lower-end regional range comparison, and arctic community spacing data (remote northern Canada / Alaska). Arctic communities are often separated by several hundred nautical miles, with limited refueling infrastructure, so an adequate range is necessary. An 800 nm range allows for connectivity between remote hubs, provides operational reserve margin, keeps the fuel fraction realistic for turboprop sizing, and avoids oversizing aircraft

structure for excessive fuel. In the sizing algorithm, range is critical because it directly determines the mission fuel fraction via Breguet equation, the gross takeoff weight, and the wing area (through W/S iteration). Too small a range underestimates fuel weight and too large a range artificially inflates takeoff weight, so it is necessary to have a realistic estimate. 800 nm balances mission realism and structural feasibility.
For our maximum payload weight, we estimated 2500lbs. To get this number we referred to the DHC-6 Twin Otter payload capacity, the Cessna 208 Caravan payload capacity, and the Raymer typical passenger weight assumptions (190–200 lb. per passenger incl. baggage). We assumed 10 passengers: 200 lb. per passenger (winter gear included) - 2,000 lbs. and 500lb for additional cargo. Arctic missions frequently mix passenger and light cargo transport. Designing strictly for passenger-only weight underestimates structural and wing loading requirements. This value drives the initial gross weight estimate, required wing area via W/S, and structural sizing trends. 2,500 lbs. is realistic and competitive relative to similar aircraft classes.

For our final parameter, we estimated the Cruise Lift-to-Drag Ratio (L/D) to be 15. This estimate came from Raymer, *Aircraft Design: A Conceptual Approach,* Typical turboprop cruise L/D range: 14–18, and ATR 42 performance data. High-wing, moderate aspect ratio STOL aircraft typically achieve L/D between 14 and 16 at cruise. Choosing 15 is conservative but realistic, prevents over-optimistic fuel fraction, and reflects external stores, landing gear fairings, and deicing hardware. This parameter directly enters the Breguet range equation. If L/D is overestimated, fuel weight becomes artificially small, which cascades into an undersized wing area and power requirements.

From our MATLAB (Appendix 3.2) code using similar aircraft's dimensions and specifications, we calculated gross takeoff mass of 5861 lb. ≈ 2658 kg, a climbing time of 12.5 min a fuel mass of 915.5 lb. ≈ 415.3 kg and a fuel volume of 136.6 gal ≈ 517 L. These estimates are close to our reference aircraft and other similar vehicles in the class such as the Pilatus PC-6 Porter which has a maximum gross takeoff weight of 6,173 lbs.

Wing Loading

The main requirements for the wing design were payload capacity, performance, range, and safety. The aircraft must carry passengers, cargo, and aerial ambulance payloads, clear a 50 ft obstacle within 800 ft, operate in Arctic conditions, and complete missions with a cruise range requirement around 400+ nm. plus loiter/reserve time. These requirements affect wing size because payload and range tend to push the design toward a larger, more efficient aircraft, while STOL performance requires low stall speed and strong low-speed lift.

The main conflict was between cruise efficiency and short takeoff performance. A higher wing loading improves cruise efficiency and reduces wing size, but it also increases stall speed and takeoff/landing distance. A lower wing loading improves STOL performance, low-speed handling, and safety on short Arctic runways, but it can increase drag and structural weight. Since the aircraft's mission depends heavily on operating from short runways, STOL capability and payload capacity were treated as the strongest design metric, while some cruise efficiency was preserved through a high-wing, moderately tapered planform and an efficient airfoil selection.

The estimated wing loading used for the aircraft is 25 lb./ ft^2. This value comes from the PM6 takeoff and landing estimate, where the aircraft weight was taken as W=6835 lb. and the selected wing loading was 25 lb./ ft^2. This gives a wing area of 273.4 ft^2. This wing loading estimate is justified because it satisfies the key mission drivers simultaneously, so they are functionally validated.

This wing area was then used in the takeoff and landing distance calculations. The resulting estimates were a takeoff distance of 658 ft and a landing distance of 769 ft, which satisfied the short-field requirement of clearing/operating within approximately 800 ft.

These values are reasonable because a wing loading of 25 lb./ft^2 is low enough to support STOL performance, but not so low that the aircraft becomes overly draggy or structurally inefficient. The selected wing area also supports the later aerodynamic model, where the aircraft used a 58 ft wingspan, 273.4 ft^2 wing area, and an aspect ratio of 12.30 for improved induced-drag performance. Overall, the selected wing loading represents a balanced

compromise between Arctic short-field capability, payload-carrying ability, and acceptable cruise efficiency.

3.3 Wing Size and Airfoil Characteristics

We estimated the wingspan of the aircraft to be 58 ft. This value came after estimating the required wing area from the aircraft's wing loading and then choosing a wing planform that fit the STOL mission. The design process identified the payload capacity and STOL capability as the dominant wing design drivers, while still trying to preserve reasonable cruise efficiency. The wing planform selected was a high-mounted, unswept, moderately tapered wing because the aircraft operates at low subsonic speeds, where sweep is not necessary and would reduce low-speed lift performance. A taper ratio of about 0.5–0.7 was selected to reduce induced drag while still maintaining predictable stall behavior. An Oswald efficiency factor of 0.80 was used as a reasonable estimate for this type of subsonic wing.

The final wing geometry used a wing area of 273.4 ft^2 and a wingspan of 58 ft, giving an aspect ratio of:

$$AR = \frac{b^2}{S} = \frac{58^2}{273.4} = 12.30$$

This aspect ratio is high for a rugged STOL aircraft, but it is reasonable because the design is trying to balance short-field performance with range and cruise efficiency. The 58 ft span also aligns well with the DHC-3 Otter reference aircraft.

For the airfoil, we selected the NACA 4412. The selection process focused on choosing a section that would provide strong, low-speed lift and forgiving stall behavior. PM5 estimated a thickness-to-chord ratio of about 0.15, with maximum thickness near 30% chord, and moderate camber of about 3–5% with maximum camber near 40% chord. These characteristics are appropriate for a STOL aircraft because they improve lift at low speeds without making the wing overly draggy.

The NACA 4412 was also chosen because it is a highly cambered airfoil that produces more lift at lower speeds than a simpler airfoil such as the NACA 2412. This is important for short

takeoff and landing on rough or unprepared Arctic surfaces. This airfoil was also noted that it has predictable stall behavior, which is important for safety during low-speed approach and landing.

The lift and drag results from PM5 gave a maximum L/D of 33.45 at α=9.75∘, a maximum lift coefficient of $C_{l,max}$ =1.4298 at α=12.5∘, and a parasitic drag coefficient of approximately C_{d0} =0.03143 at α=−0.5∘. These values are reasonable for an STOL aircraft because they show that the airfoil can generate useful lift at low speeds while still maintaining acceptable aerodynamic efficiency.

3.4 TAIL SIZING

What are the characteristics of the designed horizontal and vertical tail? Describe the airfoil selection process and the process for determining parameters as C_{HT}, S_{HT}, l_{hH}, C_{VT}, S_{VT}, and l_{Vt} for the horizontal and vertical tail, respectively. Include in the report your design values and the equations you used.

The aircraft uses a conventional tail configuration consisting of a horizontal and vertical stabilizer mounted at the aft fuselage to provide longitudinal and directional stability. A symmetric airfoil, specifically the NACA 0012, was selected for the horizontal tail because it can generate both positive and negative lift depending on the trim condition, center of gravity location, and flight phase. This is important since the horizontal tail often produces a downward force in cruise for stability, but must also provide control authority during takeoff, landing, and maneuvering.

The horizontal tail was sized using the horizontal tail volume coefficient, defined as $C_{HT} = \frac{S_{HT} l_{HT}}{S\overline{c}}$, where S_{HT} is the horizontal tail area, L_{HT} is the moment arm from the aircraft center of gravity to the horizontal tail aerodynamic center, S is the wing reference area, and $c^{-}$is the mean aerodynamic chord. Using a wing area of 273.4 ft^2, a mean aerodynamic chord of approximately 4.9 ft, a horizontal tail area of 71 ft^2, and a moment arm of 19.5 ft, the resulting horizontal tail volume coefficient is approximately 1.03. This high value is appropriate for a

STOL aircraft, as it provides strong pitch control and stability at low speeds, which is critical for short takeoff and landing operations in Arctic conditions.

Similarly, the vertical tail was sized using the vertical tail volume coefficient, defined as $C_{VT} = \frac{S_{VT} l_{VT}}{S\bar{b}}$, where S_{VT} is the vertical tail area, l_{VT} is the moment arm from the center of gravity to the vertical tail aerodynamic center, S is the wing area, and b is the wingspan. Using a vertical tail area of 32 ft^2, a moment arm of 19.5 ft, a wing area of 273.4 ft^2, and a wingspan of 58 ft, the calculated vertical tail volume coefficient is approximately 0.039. This value falls within a reasonable range for small utility aircraft and ensures sufficient directional stability and rudder authority, particularly during crosswind operations and low-speed flights on short or unprepared runways.

The selected tail sizes and parameters are justified because they balance stability, control authority, and aerodynamic efficiency while remaining consistent with the aircraft's high-wing STOL configuration and mission requirements.

3.5 TAKEOFF AND LANDING DISTANCES

In PM6, we estimated the takeoff and landing distances using simplified Nicolai-style preliminary sizing equations. We had to check to make sure our aircraft could meet the STOL requirement of operating within approximately 800 ft while carrying the design weight.

The aircraft weight was taken as 6835 lb, and the selected wing loading was 25 lb/ft^2, which gave a wing area of:

$$S = \frac{W}{\frac{W}{S}} = \frac{6835}{25} = 273.4\, ft^2$$

For takeoff, the stall speed was first calculated using:

$$V_{S,TO} = \sqrt{\frac{2W}{\rho S C_{Lmax,TO}}}$$

Where W is aircraft weight, ρ is air density, S is wing area, and $C_{Lmax,TO}$ is the maximum lift coefficient in takeoff configuration. The takeoff speed was then estimated as:

$$V_{TO} = 1.2V_{S,TO}$$

The approximate thrust at takeoff speed was calculated from the propeller power relation:

$$T = \frac{\eta_p P}{V_{TO}}$$

where η_p is propeller efficiency and P is engine power. The thrust-to-weight ratio was then:

$$\frac{T}{W}$$

Finally, the takeoff distance was estimated using:

$$S_{TO} = \frac{20.9(W/S)}{\sigma C_{Lmax,TO}(T/W)}$$

For landing, the landing stall speed was calculated as:

$$LV_{S,L} = \sqrt{\frac{2W}{\rho S C_{Lmax,L}}}$$

and the approach speed was estimated as:

$$V_A = 1.3V_{S,L}$$

The landing distance was then estimated using:

$$S_L = \frac{80(W/S)}{\sigma C_{Lmax,L}}$$

Using $W = 6835lb$, $W/S = 25$ lb/ft^2, $S = 273.4$ ft^2, $\eta_p = 0.85$, $P = 620hp$, $C_{Lmax,TO} = 2.2$, $and\ C_{Lmax,L} = 2.6$, the calculated results were a takeoff distance of 658 ft and a landing

distance of 769 ft. These values meet the competition/STOL requirement because both are below the required 800 ft field length target. This shows that the selected wing loading, wing area, power level, and high-lift assumptions are reasonable for the aircraft's intended Arctic short-field mission.

3.6 AIRPLANE'S C_{D0} AND DRAG POLAR DETERMINATION

Determination of C_{D0}

To begin the determination of our parasitic drag (zero-lift drag coefficient), we need to sum the drag contributions from various parts of the aircraft. For our design, we will need to sum the drag from the fuselage, wing, tail, and landing gear. Starting with the fuselage, we can use the following relation to estimate the fuselage zero-lift coefficient of drag:

$$C_{D0,f} = C_f \left(\frac{S_w}{S_0}\right)\left[1 + \frac{60}{\left(\frac{l}{d}\right)^3} + 0.0025\left(\frac{l}{d}\right)\right]$$

Where C_f is the baseline skin friction, $\frac{S_w}{S_0}$ scales the drag to the reference area, and $\frac{l}{d}$ is the fineness ratio. We can then expand the equation to achieve the full formula that we will employ:

$$C_{D0,f} = \left[C_f \left(\frac{S_w}{S_0}\right)\left[1 + \frac{60}{\left(\frac{l}{d}\right)^3} + 0.0025\left(\frac{l}{d}\right)\right] + 0.07C_{Db}\right] \cdot \frac{S_w}{S_{ref}}$$

We can compute skin friction with the following formula:

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}}$$

Where Re is the Reynolds number. We can then obtain the base drag term with:

$$C_{D,b} = 0.029\left(\frac{d_b}{d}\right)^3$$

Where d_b is the base diameter of the fuselage. We can then begin to estimate the next parameter, which is the tail drag. We can use the following formula to estimate the tail drag:

$$C_{D0,tail} = \frac{C_f \cdot FF \cdot S_{wet,tail}}{S_{ref}}$$

Where FF is the form factor of the tail. Finally, we can use a fixed value of 0.02 to estimate the drag contribution of a fixed landing gear. Since we have initially assumed that we will have a fixed landing gear, we include the value here.

Using a MATLAB script (Appendix 3.6) we can estimate the zero-lift coefficient of drag along with the contribution areas from each part of the aircraft. The script results in a zero-lift coefficient of 0.0283. We also obtain the following contribution areas:

Table 1. Zero-lift Coefficient Contribution Areas

Component	**Area [ft^2]**
Wing	2.05
Fuselage	2.475
Horizontal Stabilizer	0.537
Vertical Stabilizer	0.454
Nacelle	0.421
Landing Gear	1.8

This leaves us with a total contribution area of 7.737 square feet.

Determination of Drag Polar

To determine the standard form of our drag polar, we will need to calculate the induced drag coefficient. The standard form of the drag polar is given as:

$$C_D = C_{D,0} + kC_L^2$$

Where the induced drag is given as:

$$k = \frac{1}{\pi e AR}$$

Where e is the Oswald efficiency factor and AR is the aspect ratio. Using our previously estimated wingspan and wing surface area, we can calculate the aspect ratio to be:

$$AR = \frac{b^2}{S} = 12.31$$

We also assume that our Oswald efficiency factor is 0.8. Plugging all our known values in: we obtain a drag polar for our aircraft defined as:

$$C_D = 0.032 + 0.0323C_L^2$$

Using a MATLAB script (Appendix 3.6) we can calculate this and plot the drag polar. The drag polar can also be found in section 4.4 of this report.

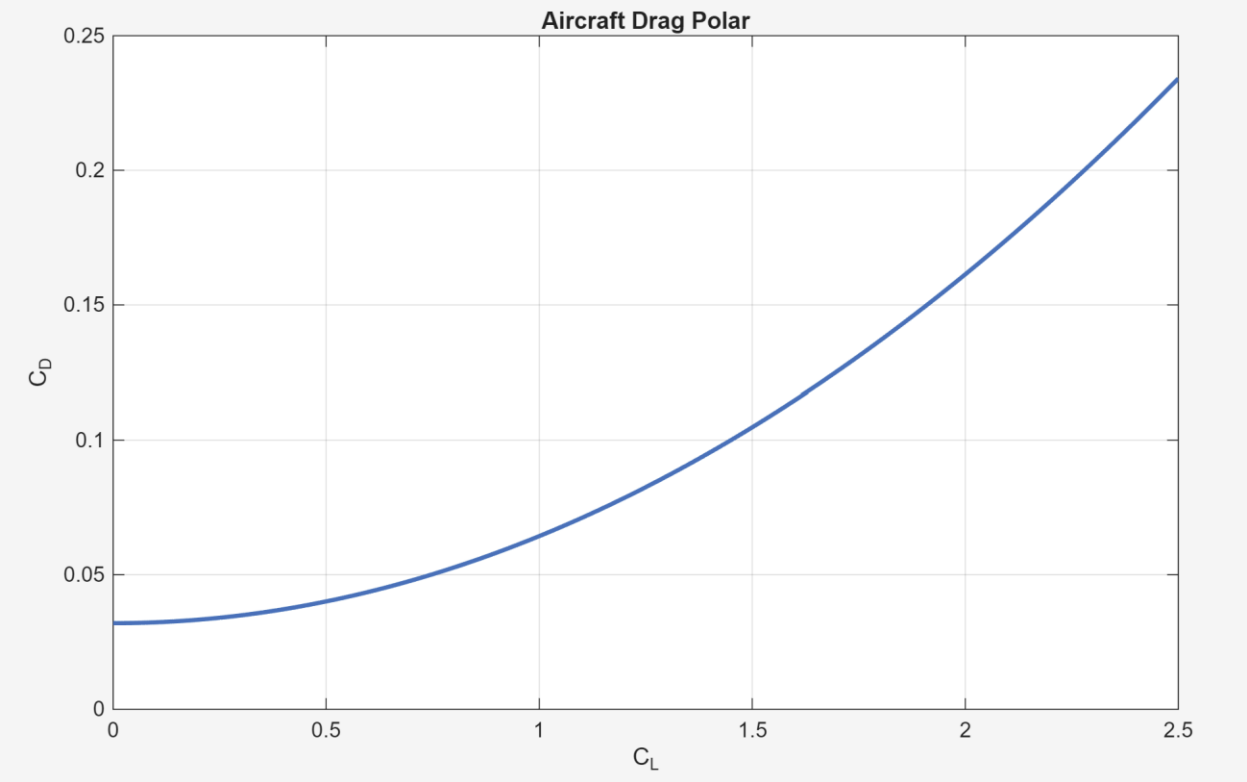


Figure 9. Plot of Aircraft Drag Polar

4 Detailed Concept, Selection, and Performance Analysis

In this section, you will start to see the performance impact of our estimated design parameters and how they begin to fit together to create an actual aircraft. You will see the concept generation process and the final form of our aircraft.

4.1 Concept Generation

Using our initial assumptions and some of our design parameters, we were able to generate three different concepts for our aircraft. While the body of the aircraft remains the same, we change the locations of certain features such as the wings and engines. We also switch between a single engine and two engines in the generated concepts. Below are the three design sketches that we generated:

Design Concept 1

Our first design concept consisted of an overwing design with a nose mounted engine. We chose to go with an overwing design in this concept generation to allow for the most space for ground-crew to work. Since all the mission objectives involve transportation, we wanted to make sure that this aircraft's cabin was easily accessible. The nose mounted engine also allows for extra maneuvering space on the ground. In many existing cases, a single nose mounted engine is more than enough to supply sufficient power for an STOL aircraft.

We also chose to include fixed landing gear to increase simplicity. A fixed landing gear will require less maintenance as there are far fewer moving parts. This decrease in maintenance will also lead to lower maintenance costs. Although the costs of maintenance are beyond the scope of this report, we find it worth mentioning in our design considerations. Below is an illustrated hand-sketch of this design concept from our preliminary design generation.

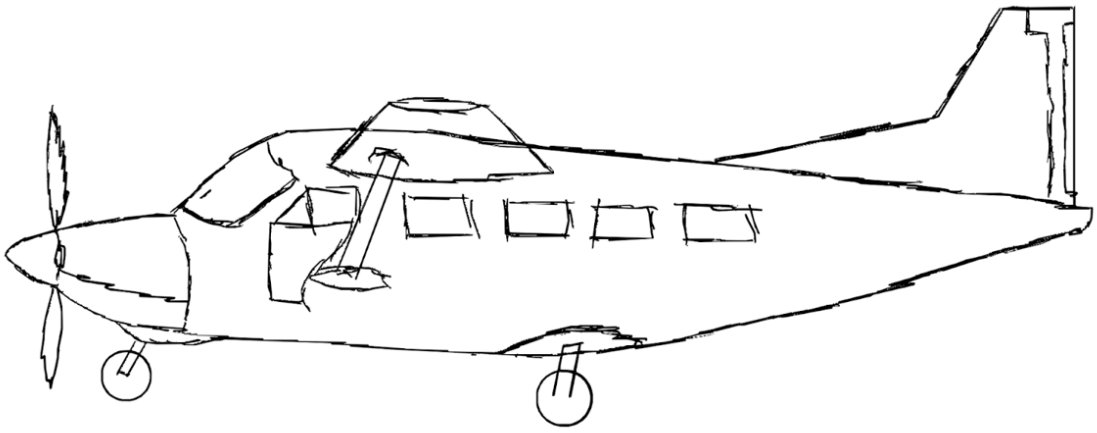


Figure 10. Overwing Design Concept with Nose Mounted Engine

Design Concept 2

For our second design concept, we moved to an underwing design while keeping the nose mounted engine. An underwing design can allow for more structural loading throughout the wing and can be an advantage in the case of an emergency landing [Pilot Institute]. This design concept includes a retractable landing gear to reduce drag. In the event of a landing

gear deployment failure, the underwing can help keep the main fuselage above the wing and ease the landing. This would not be available to an overwing design with retractable landing gear. An underwing design can also offer easier access to internal components and fuel tanks that are located inside the wing. Below is an illustrated hand-sketch of our second design concept.

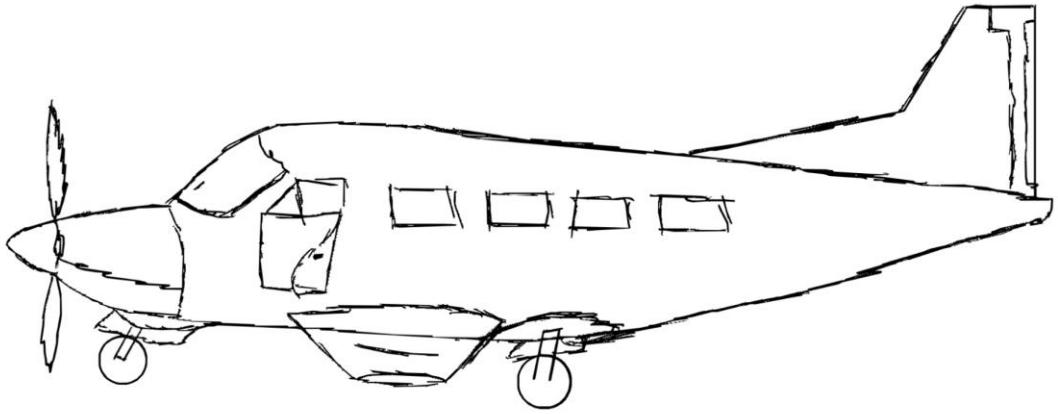


Figure 11. Underwing Design Concept with Nose Mounted Engine

Design Concept 3

Design concept 3 features an overwing design with a fixed landing gear like design concept 1. The difference in this design concept is that there are now two engines mounted to the wing. We believe it to be important to include a design that features two engines to compare it to that of a single engine. Using a twin-engine design gives the aircraft more available power than that of a single engine. We would be able to carry more payload while maintaining a similar range.

Using a twin-engine design would decrease the amount of space the ground-crew has to work with, and maintenance would be much higher. While not much of an increase, we could expect to see slightly higher specific fuel consumptions while using two engines [Pilot Institute]. Below is an illustration of our third design concept.

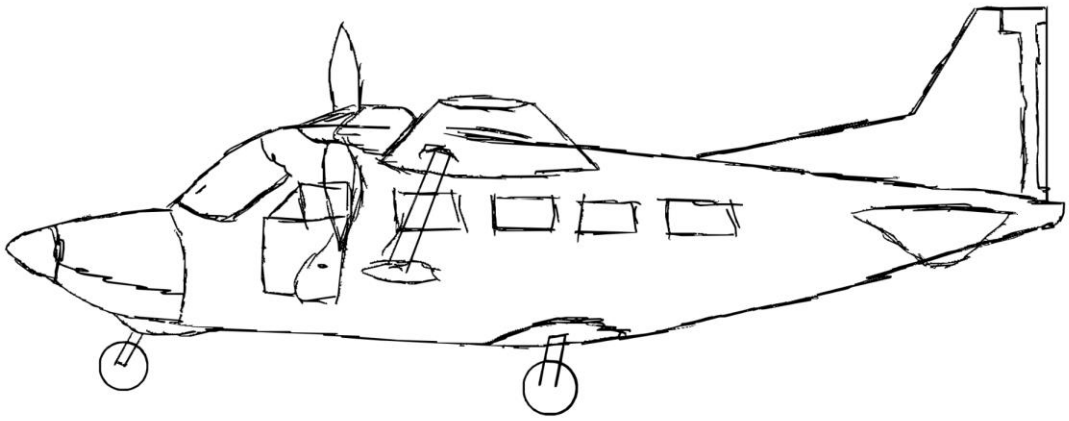


Figure 12. Overwing Design Concept with Twin Engines

4.2 CONCEPT SELECTION

Concept Requirements

Based on the requirements that we identified in section 2.2, we have identified the following requirements of the most significance:

- Payload Capacity
- Takeoff Distance
- Engine Power
- Range
- Efficiency
- Ease of Use
- Destination Turnaround

We will use these requirements to judge our design concepts and select the one that meets all our criteria and meets the requirements of the missions.

Concept Selection

The design concept that we have selected is our first design concept. Our aircraft will feature an overwing design with a single, nose mounted turboprop engine. The aircraft will also feature fixed landing gear.

Our aircraft will feature a single engine to save weight and to increase efficiency. A single engine will reduce the overall maintenance of the aircraft and allow for increased pilot visibility. Modern turboprop engines are plenty powerful and efficient for the scope of this project and eliminated the need for a twin-engine design. Later, we will go into depth concerning the specifics of the engine.

We have also gone with an overwing design like what is currently used on the Cessna Caravan. This overing design will allow for the ease of transportation and loading the aircraft with the required cargo. Overall access to the aircraft was very important to us and can be severely limited with an underwing design. This is especially important for the medical transport mission where time is critical, and time cannot be wasted loading the patient into the aircraft.

The fixed landing gear design will also reduce the maintenance of the aircraft and allow for more structural rigidity. This will allow us to land on slightly rougher terrain and is quite common on existing back-country aircraft. The reduction of parts will also reduce weight and allow us to carry more payload, which is the most important design consideration.

Considering the cargo loading, we will also have a panel that will act as a cargo loading door that can accommodate medical transport, cargo transport, and passenger transport. This will also allow for quick cabin swaps based on the required mission.

Below is a digitized concept sketch of our design selection:

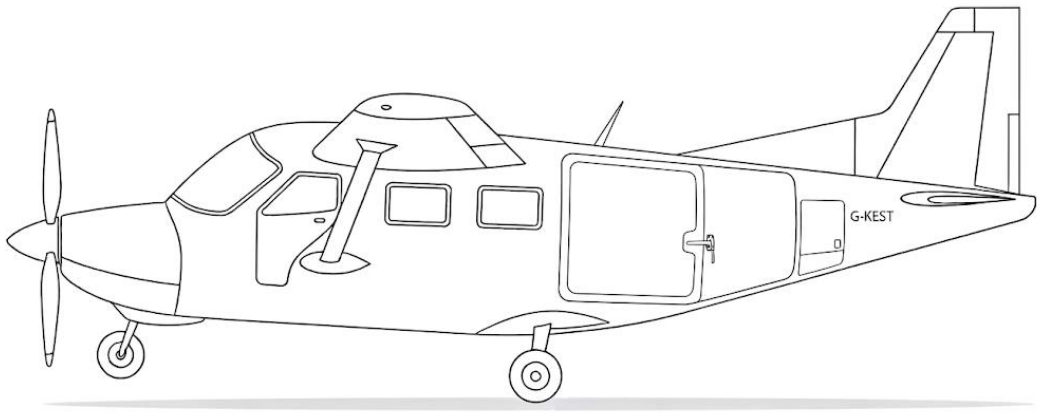


Figure 13. Digitized Concept Design [Gemini]

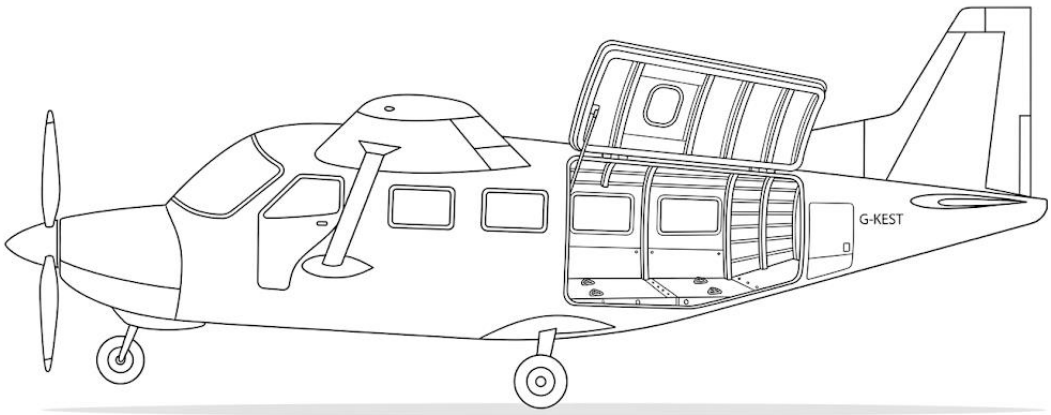


Figure 14. Digitized Concept Design with Cargo Door Open [Gemini]

4.3 Concept Refinement

Main Wing Selection

For our aircraft design, we chose the NACA 4412 airfoil. Figure 15 shows the cross-sectional geometry of this airfoil. We wanted our airfoil selection to utilize camber, and this airfoil possesses a maximum camber that is 4% of the chord. Then from the rest of the naming convention (4412), the airfoil has a maximum thickness of 12% of the chord, and the location of the maximum camber is at 40% of the chord from the leading edge.

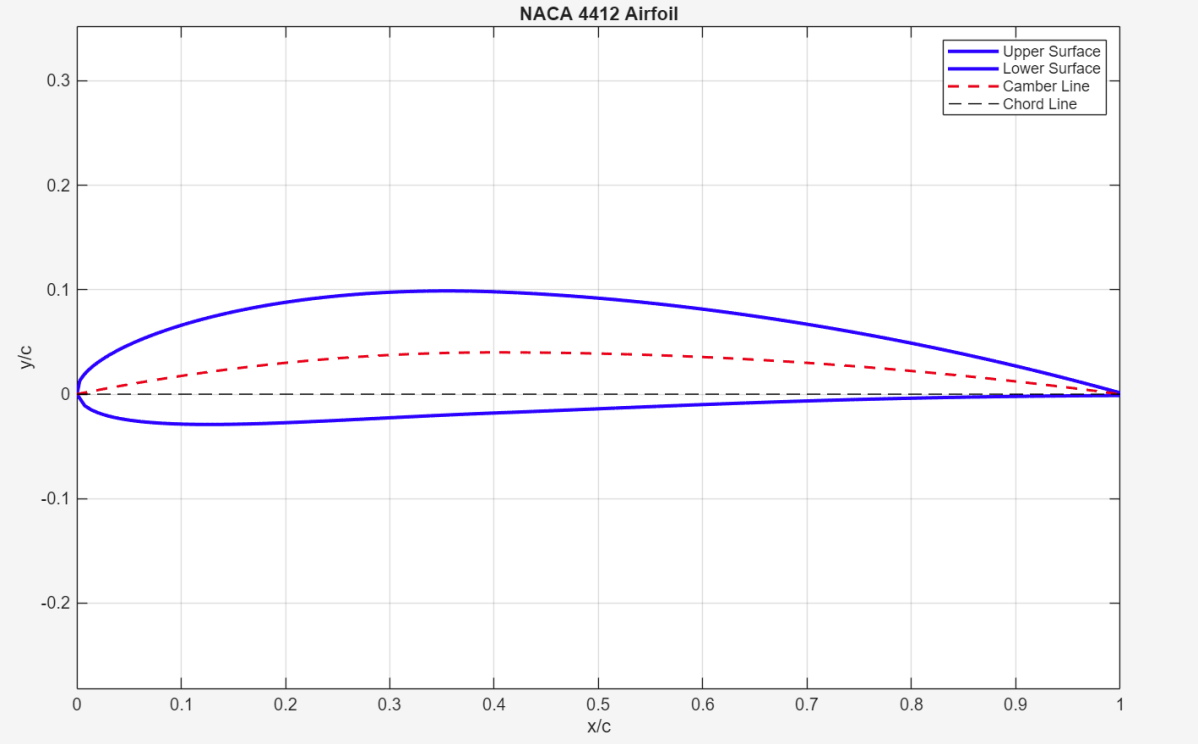


Figure 15. NACA 4412 Airfoil Cross-Section

We chose this airfoil because of its lower stall speed and suitability for short takeoff and landing aircraft. The NACA 4412 airfoil also has a gentle stall behavior because of its slight camber. This will make it easy to control the aircraft at slower speeds and allow for a shorter landing distance. When combined with flaps and slats, this airfoil can produce extremely high lift coefficients which is ideal for a STOL aircraft.

Compared to a symmetric airfoil, a cambered airfoil is far more suited for an STOL aircraft. While a symmetric airfoil is excellent for cruise speed efficiency because of its extremely low drag, the stall speed will be much higher than that of a cambered airfoil. Our goal was to balance the trade-offs between high low-speed performance and acceptable cruise speed efficiency. A suitable comparison to the NACA 4412 airfoil is the NACA 0012 airfoil which is a symmetric airfoil.

The airfoil's 12% thickness also gives us enough room to include any reinforcements the wing may need and can carry fuel. However, with an airfoil with camber, there will be tradeoffs. With the addition of more camber, the aircraft will experience a higher drag during cruise because of the more turbulent airflow. This will decrease the cruise speed efficiently, but for the purpose of this aircraft we have deemed this a worthy trade-off.

For the icing problems, there are several design approaches that we could take to mitigate the problem. Existing STOL aircraft feature different wing deicing systems such as an inflatable boot, electric heating, or hot bleed air from the aircraft's engine. For our aircraft, the system that makes the most sense is an inflatable rubber bladder that can crack the ice on the leading edge of the aircraft. This option is energy efficient and can save power to be used elsewhere on the aircraft.

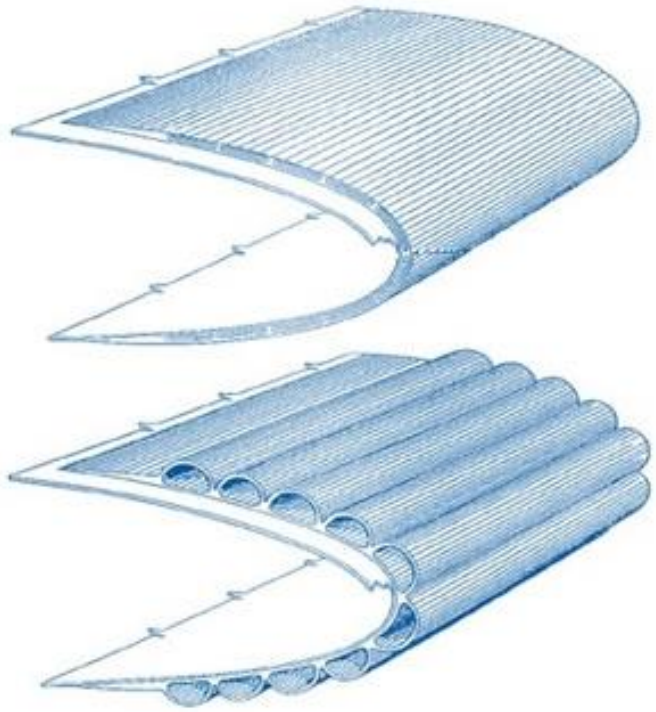


Figure 16. Inflatable Leading Edge for Deicing [icing]

Stabilizer Wing Selection

For our stabilizer design, our aircraft will use a symmetric airfoil because we will want to produce lift in whatever direction we need it given the state of the control inputs from the pilot. Unlike a non-symmetric airfoil, such as a NACA 2412 or 4412, the NACA 0012 airfoil will have zero lift in its neutral position. Using an airfoil like the NACA 0012 for our horizontal stabilizer will provide suitable stability and performance for our STOL aircraft design.

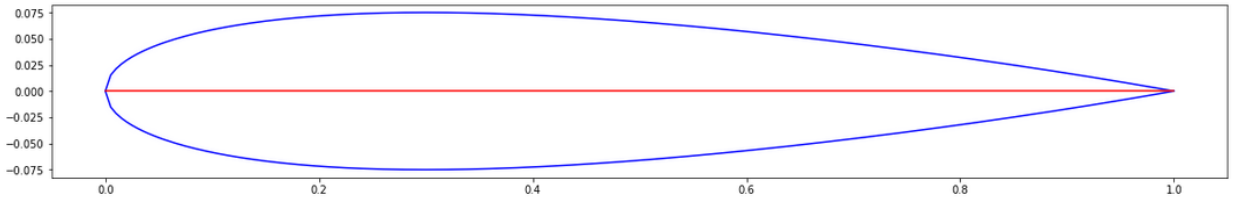


Figure 17. NACA 0012 Airfoil Cross Section

Landing Gear

Indicated in our concept selection in the previous section, we have chosen to equip our aircraft with a fixed landing gear. Our goal was to use a simplistic landing gear design to conserve weight and to reduce overall maintenance. A fixed landing gear does both things to perfection and reduces the number of failure points on the aircraft.

A large disadvantage to retractable landing gear is that it is a complex mechanism. A retractable landing gear requires many moving parts and it to successfully lock into place on every landing sequence. A fixed landing gear always remains down, and there is next to no risk of mechanical failure. A fixed landing gear is also another checklist item that can be removed for the pilot, meaning fewer items for the pilots to remember.

Our aircraft design will feature a tricycle gear design (also known as a delta gear configuration) for simplicity and pilot ease. This configuration consists of two main wheels that are located under the fuselage and a third wheel under the nose of the aircraft [J.A Air Center]. The tricycle landing gear configuration is a tried and tested configuration that features many existing short takeoff and landing aircraft. Below is a diagram of the tricycle landing gear configuration.

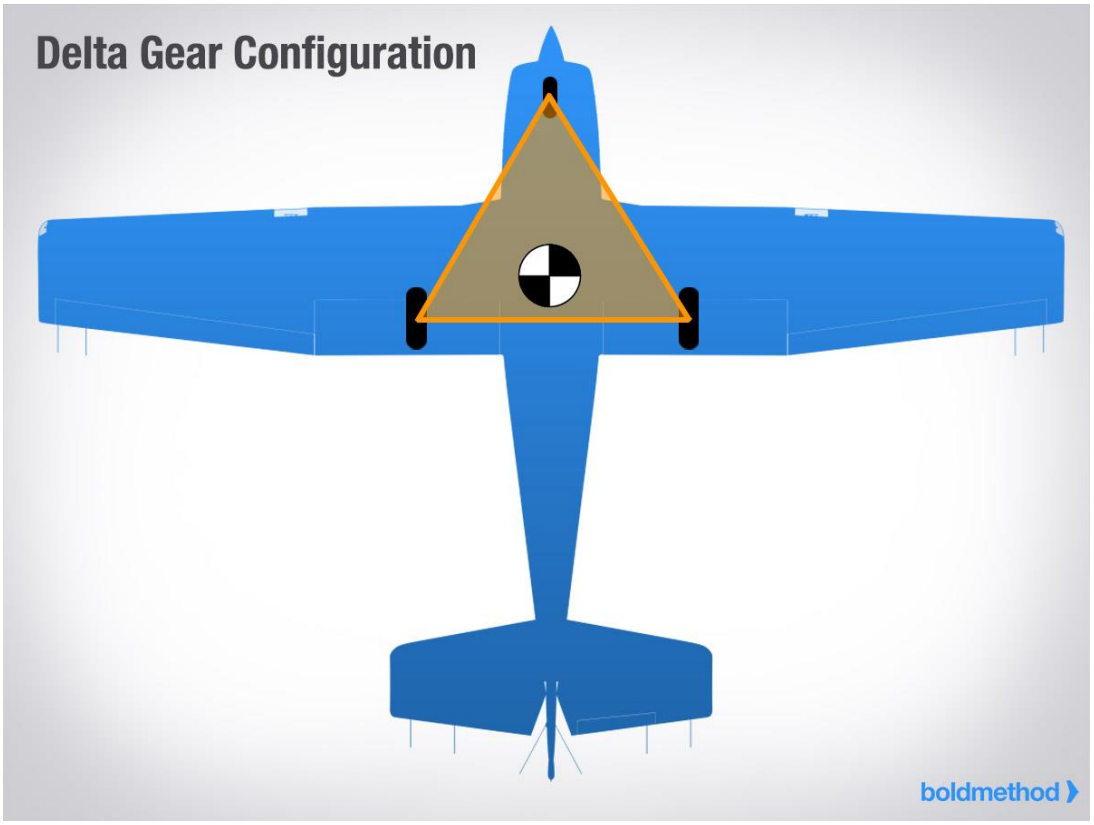


Figure 18. Tricycle (Delta) Landing Gear Configuration [Boldmethod]

Another quite common landing gear configuration is the tailwheel configuration. This design features two main wheels toward the front of the aircraft and one smaller wheel at the very rear of the aircraft. Compared to the tricycle configuration, the tailwheel design is much more unstable because the center of gravity is farther from the main gear than in the tricycle configuration [Boldmethod].

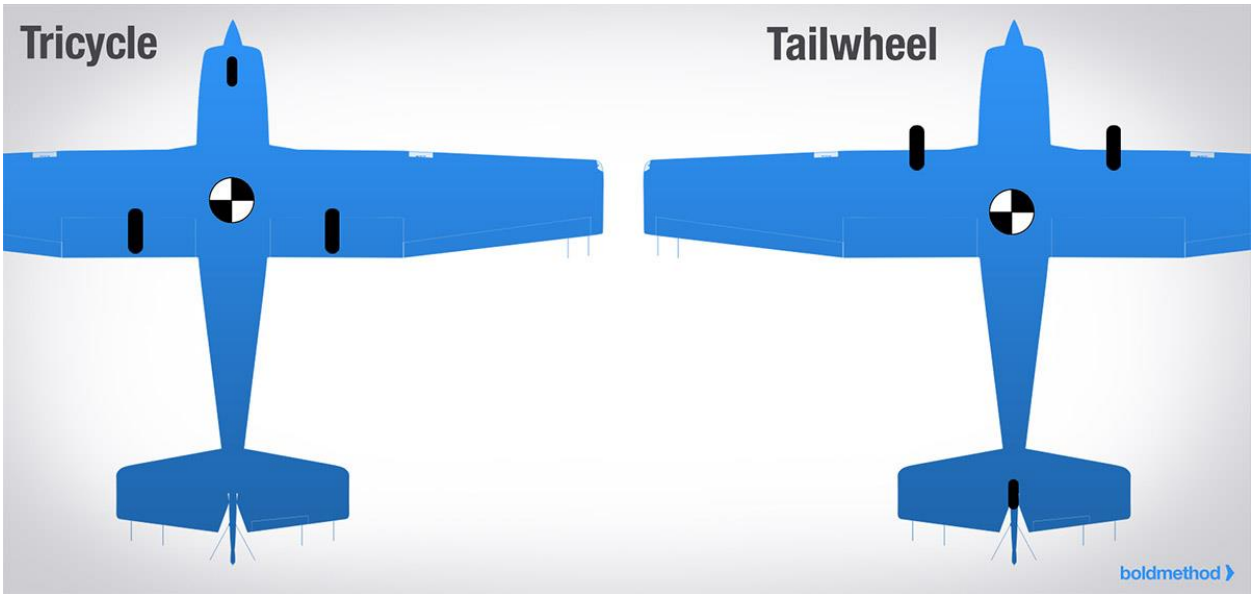


Figure 19. Tricycle & Tailwheel CG Comparison [Boldmethod]

Since our aircraft is going to be operating in less than favorable conditions, we select the landing gear configuration that is going to give us more stability, especially when operating on ice.

This then raises the question of icing and how we can mitigate it with fixed landing gear. Since the landing gear structure itself is not a critical surface such as the wings or vertical and horizontal stabilizers, we do not have to heat these components. To battle the icing problem that will be present in Arctic conditions, we will need either an electric heater near the brakes, or we can route hot bleed air from the engine to warm the brakes and prevent icing. An existing aircraft, the Hawker Beechcraft King Air, uses a brake deicing system such as this and is proven to be extremely effective at deicing. Commonly, other aircraft subsystems use engine bleed air, so it will be important to ensure the pilots are aware that they cannot use this deicing system outside of takeoff and landing. Since landing gear is not necessary for much else, this will be an effective solution to the icing problem in Arctic conditions.



Figure 20. Brake Deicing Hose for Engine Bleed Air [AOPA]

Engine Selection

Building from our initial assumptions outlined in section 1, our aircraft will be designed with a turboprop engine. For a smaller STOL aircraft such as ours, a jet engine does not make sense. While a jet engine may produce more thrust and allow for a higher cruising altitude, the operating costs and runway requirements would be too significant to use a jet engine. A turboprop engine offers better fuel efficiency and superior low-speed handling. Following concept generation, we identified three turboprop engines that would have the potential to power our aircraft. Below is a table outlining the parameters of each engine.

Table 2. Engine Parameters (Maximum Continuous Power)

Engine	**Shaft Power [SHP/kW]**	**Output Speed [rpm]**	**Dry Weight [lbs.]**
Pratt & Whitney PT6A-140	867/647	1900	417
Honeywell TPE331-43A	575/429	2000	336
GE Catalyst	1,300/970	1700	639.7

Based on what we have learned about each of the identified engines, we have elected to equip our aircraft with the Pratt & Whitney PT6A-140 turboprop engine. We have selected this engine because of its balance and performance for STOL operations. The PT6A has strong low-speed torque and is very well suited for short takeoffs. This engine is also featured on existing STOL aircraft such as the Twin Otter and PC-12. Compared to the other engines, it has a nice weight balance and will allow for more payload than the GE Catalyst. The PT6A will also supply more power than the Honeywell TPE331, which is needed to take off at the required distance.

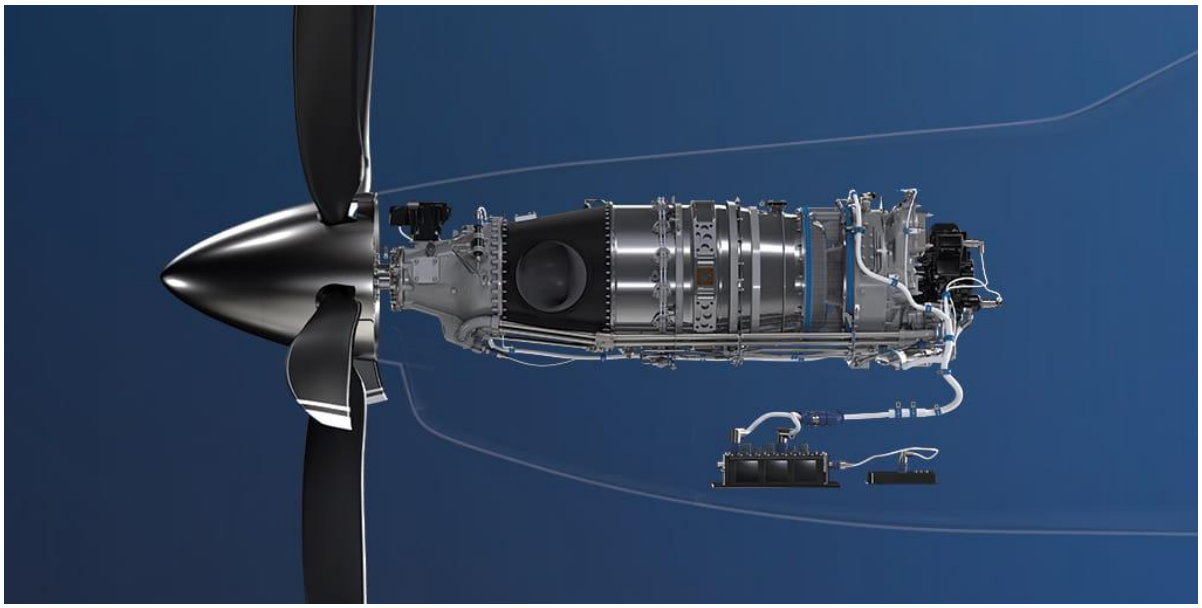


Figure 21. Pratt & Whitney PT6A-140 Turboprop Engine [Google Images]

Considering the Arctic conditions that this engine will be performing in, it makes the PT6A-140 an extraordinarily strong choice for our aircraft. Frigid air is naturally denser than warm air, which means that our engine selection will benefit from a higher mass flow through the compressor and more efficient combustion. However, we will still need to mitigate icing concerns. The PT6A-140 comes with inlet anti-ice inlet, but we can still see power loss from ice buildup. To manage the icing in the Arctic conditions, we will need to preheat the engine to warm the engine oil and melt the ice. This will allow for a proper engine to warm up and optimal performance in cold conditions.

The PT6A-140 utilizes a backwards airflow design to produce power and thrust. Air enters the rear of the engine where it is compressed and combusted to power the propeller. Below is a diagram of the engine showing the flow of air from the first stage to the last stage:

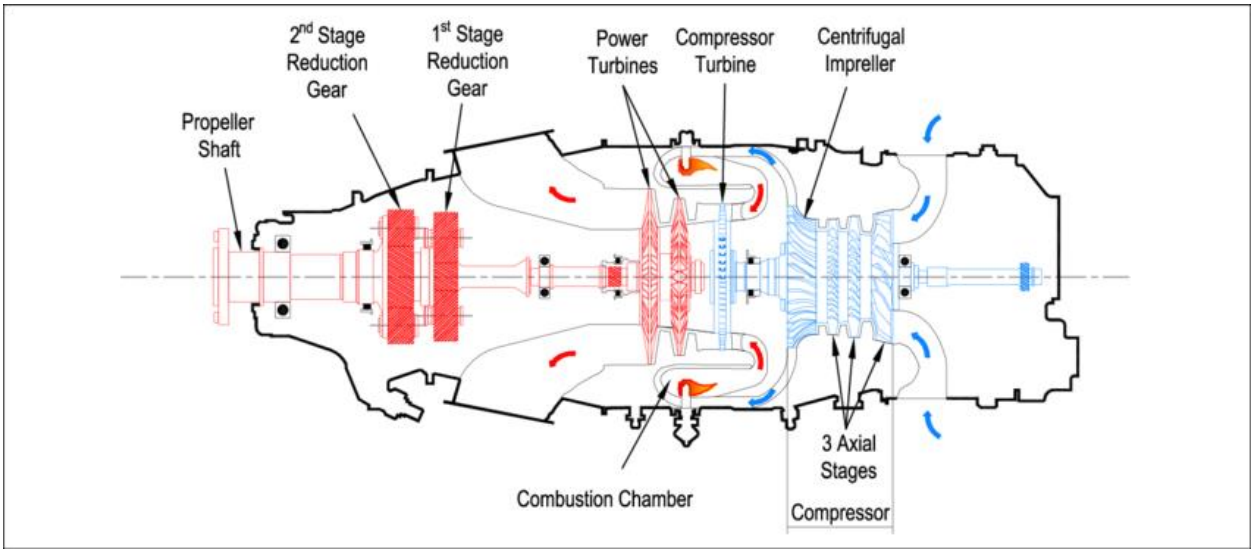


Figure 22. Pratt & Whitney PT6A-140 Engine Stage Diagram [Camilo Bayona-Roa et al., 2019]

This engine design is modern and current, making it perfect selection for our aircraft as support is common and easy to obtain. This will ease maintenance and maintenance time as there are plenty of parts and mechanics that can operate on the engine. This will increase maintenance turnaround time compared to an engine like the GE Catalyst, which is an extremely new engine and may not be easily serviceable as parts are scarce, and few mechanics know how to work on it. This reinforces our decision to use the PT6A-140 engine as scarcity and knowledge of the engine are not a problem.

Dimensioned Drawings

Using our previously calculated design parameters, we can create a dimensioned drawing of our chosen design. Below is the dimensioned drawing and an expanded and enlarged table containing the specifications.

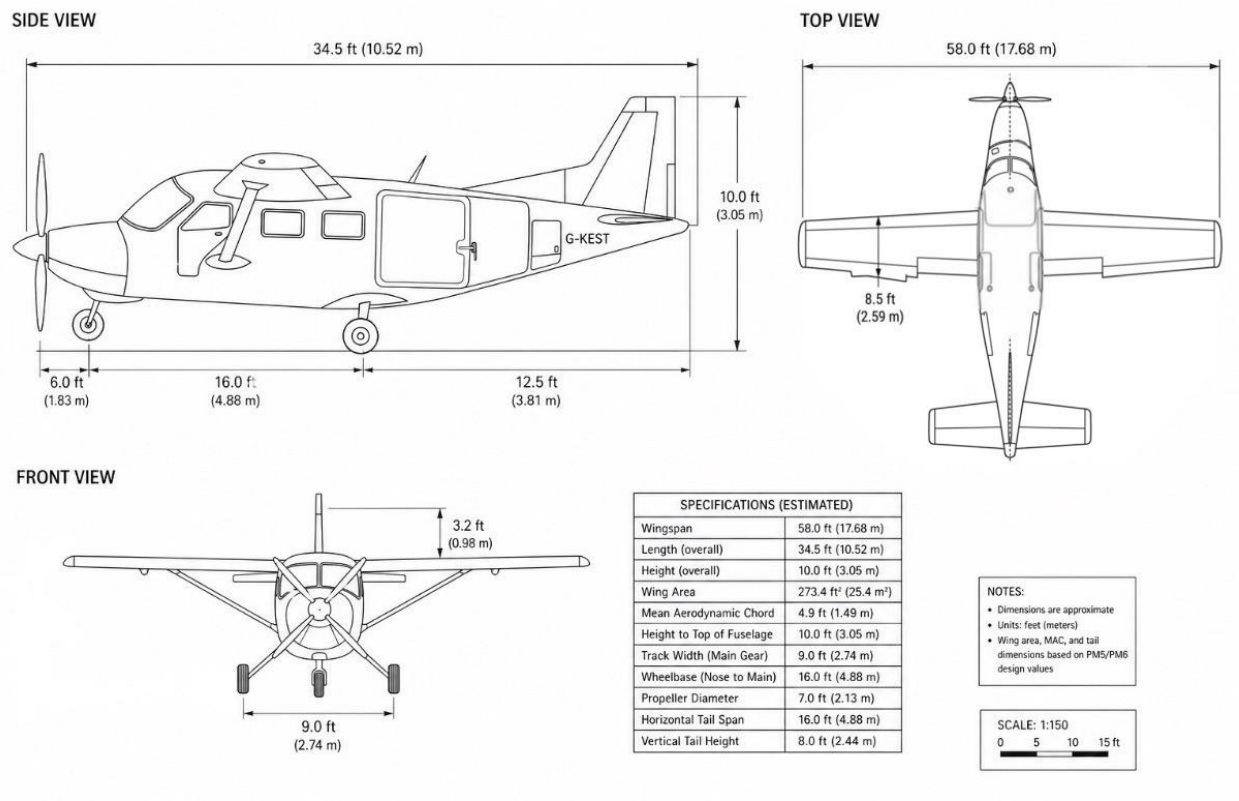


Figure 23. Dimensioned Drawing

Table 3. Aircraft Design Specifications

SPECIFICATIONS (ESTIMATED)	
Wingspan	58 ft
Length (overall)	34.5 ft
Height (overall)	10 ft
Wing Area	$273.4 ft^2$
Mean Aerodynamic Chord	4.9 ft
Track Width	9 ft

Wheelbase (Nose to main)	16 ft
Propeller Diameter	7 ft
Horizontal Tail Span	16 ft
Vertical Tail Height	8 ft

Misson Specific Considerations

Now we will need to consider the type of systems and avionics that our aircraft will require to remain operational in all types of conditions. First, we will consider the types of systems that are required for our aircraft to be operational.

- Flight Controls
- Electrical Power System
    - Redundant Systems
- Environmental System
    - Air Conditioning
    - Heating System
    - Oxygen System
    - Insulation
- Fire Suppression System
    - Engine Extinguisher Bottles (Multiple for Redundancy)
    - Fire Suppression for Cabin (Fire Extinguisher)
- Hydraulic System
    - Flaps
    - Redundant Systems
- Exterior Lights
    - Position Lights (Red & Green Wing Lights)
    - Anti-Collision Lights (Top of Fuselage, Strobe)

    - Landing Lights
    - Taxi Lights
- Cockpit Voice Recorder

These systems will be critical for all types of missions and the aircraft being operational. For many of the subsystems, we will implement at least two systems to maintain redundancy throughout the aircraft if a primary system fails. This will reduce the risk of losing a system for good and rendering the aircraft un-operational.

We also need to consider the avionics that our aircraft will need to be able to operate in continuous daylight or night conditions. Using the guidelines set forth in FAA 14 CFR Part 23, we have identified the necessary avionics for our aircraft:

- Airspeed Indicator
- Altimeter
- Compass/Heading Indicator
- Vertical Speed Indicator
- Navigation Equipment
    - Charts
    - VHF Receiver
    - VOR Receiver
    - ILS Receiver
    - Autopilot
    - Digital Avionics Systems
    - GPS
- Radio
    - VHF Radio
        - 8.33 kHz Spacing (For FL200)
    - HF Radio
    - Transponder (ADS-B Out)
- Weather
    - Outside Temperatures

- Outside Pressures
- Storm Radar

These instruments will be necessary to carry out the transport missions that will be required for our aircraft. While specific instrumentation is outside the scope of our design objectives, we would have to consider using a fully digital system (Glass Cockpit) or a mix of steam gauges and a digital system. An example of each can be found in the below figures.



Figure 24. Glass Cockpit [Google Images]



Figure 25. Combination of Digital and Analog Avionics [Google Images]

Now considering our specific mission types, we will need to identify the necessary equipment required for the cargo mission variant. We have identified what we believe will be necessary to complete the cargo mission.

- Modular Floor
- Tie Downs
    - Straps
    - Rachets
    - Spacing Bars
- Cargo Netting

The modular floor is going to be especially important because it will allow us to quickly swap interior components to accommodate the mission variant that the aircraft will be executing. The modular floor will allow the cargo to snap into place and make it easier to move the cargo around. The figure below illustrates this modular floor.



Figure 26. Modular Cargo Flooring [aerosavvy]

Using this system, we will be able to easily swap to the passenger transport mission in which we will need the following supplies:

- Seats
- Luggage Space
- Oxygen System
- Cargo Netting

We can then also easily switch to the medical transport mission, in which we will need:

- Stretcher
- Additional Power for Medical Equipment
- Supplemental Oxygen
- Medical Supplies
- Seats

This modular cargo floor is what is going to significantly reduce the turnaround time at different airstrips when the mission type needs to be altered. Ease of use for the ground-crew will be paramount, especially in the medical transport mission when the patient could be time critical. This system will allow seats and cargo to be removed quickly, while still ensuring proper safety.

4.4 Performance Analysis

Drag Polar

From section 3.6 we previously found that the zero-lift coefficient of drag is 0.032 and the drag polar is:

$$C_D = 0.032 + 0.0323C_L^2$$

We can then use a MATLAB (Appendix 4.4) script to plot the coefficient of drag as a function of the coefficient of lift.

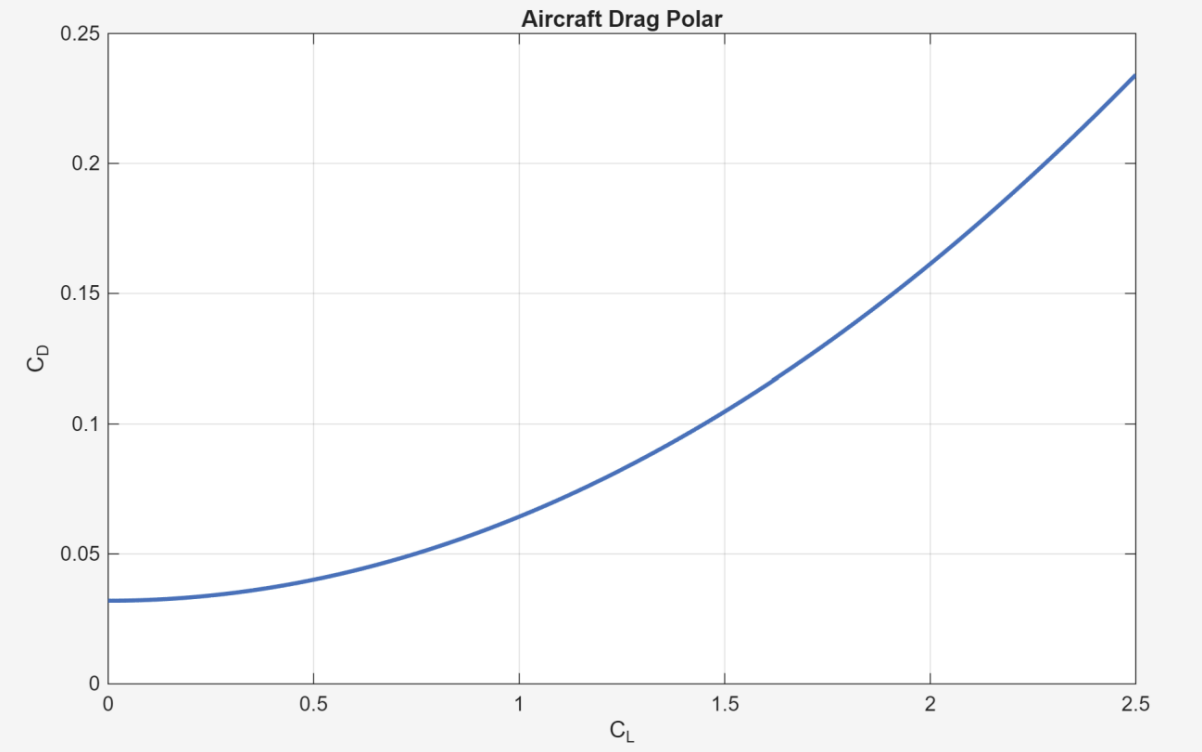


Figure 27. Plot of Aircraft Drag Polar

C_D vs. Flight Speed

Using our previously obtained drag polar equation, we can use this to plot our coefficient of drag as a function of the flight speed. Using our drag polar and keeping the constant as symbols, such as C_{D0} for the zero-lift drag and k for the induced drag we obtain:

$$C_D = C_{D0} + kC_L^2$$

Given the assumption of sustained level unaccelerated flight (SLUF) we can treat the lift as equal to the weight of the aircraft. This allows the transformation:

$$L = \frac{1}{2}\rho V^2 S C_L \rightarrow C_L = \frac{2W}{\rho V^2 S}$$

Where ρ is the air density, V is the velocity, S is the wing surface area, W is the weight of the aircraft, and C_L is the coefficient of lift. Now that we have obtained the coefficient of lift as a function of the velocity of the aircraft, we can plug this into our obtained drag polar to obtain the coefficient of drag as a function of velocity.

$$C_D = C_{D0} + \frac{2kW^2}{\rho^2 V^4 S^2}$$

Using a MATLAB (Appendix 4.4) script, we can use this equation along with our drag polar parameters and the following assumptions:

- Imperial Units
- Air density for 20,000 feet: 0.001267 $\frac{slugs}{ft^3}$
- Aircraft weight: 6835 pounds
- Wing area: 273.4 ft^2
- Velocity varies from 100 to 600 $\frac{ft}{s}$

We get the resulting plot:

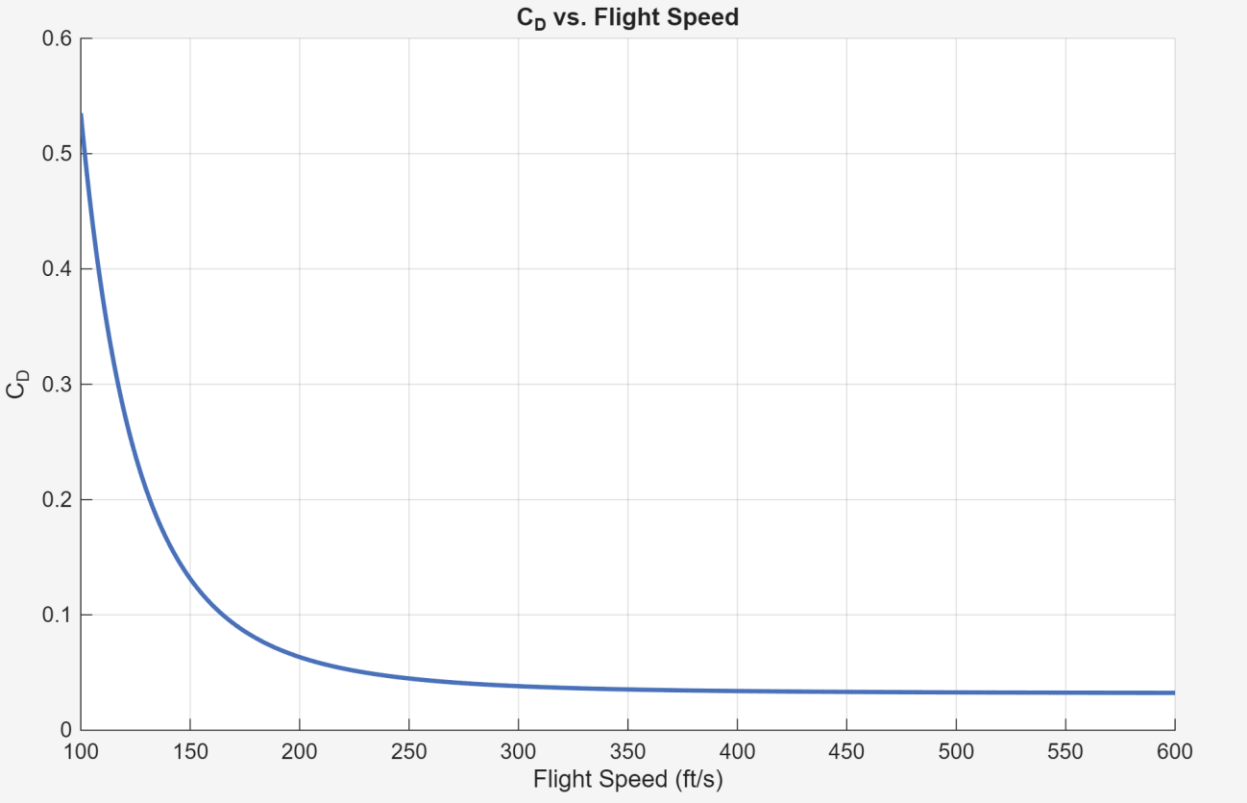


Figure 28. C_D vs. Flight Speed

The resulting curve shows asymptotic behavior as flight speed increases. We see this behavior because the induced drag is inversely proportional to the flight speed. We can use the following relationship to define this behavior:

$$\lim_{V \to \infty} C_D(V) = C_{D0}$$

Based on our drag polar and the resulting relationship between flight speed and the coefficient of drag, flying at a faster flight speed will significantly reduce the overall drag of the aircraft.

Maximum Range

Using the design parameters previously stated in this design report, we can now identify the maximum range of our aircraft using the Breguet Range Equation. The Breguet Range equation for a propeller aircraft is defined as:

$$R = \frac{\eta_p}{c_p}\left(\frac{L}{D}\right) ln\left(\frac{W_i}{W_f}\right)$$

Where η_p is propeller efficiency, c_p is the power specific fuel consumption, $\frac{L}{D}$ is the lift to drag ratio, W_i is the initial weight of the aircraft, and W_f is the final weight of the aircraft. We will assume that the aircraft is in the cruise phase for the entirety of the flight to match the preceding assumptions that have been made. Using this, we obtain the following parameters for the equation:

- $\eta_p = 0.8$
- $c_p = 1.2 \times 10^{-6}\ s^{-1}$
- $W_i = 2658\ kg$
- $W_f = 2242.7\ kg$
- $\frac{L}{D} = 15$

Note that for the simplicity of calculation the units have been converted from imperial to metric

Plugging in these parameters to the Breguet Range Equation for a propeller aircraft, we obtain a range of approximately 850 nautical miles. This value is consistent with the design decisions that we have been making and is right on target with our goals of the aircraft.

With a range of 850 nautical miles, our aircraft will be very practical for flying Arctic missions and supporting many communities while also being able to reach farther bases located outside of the Arctic Circle.

Maximum Endurance

Similarly, we can use the Breguet Endurance Equation to calculate how long the aircraft can stay airborne. The Breguet Endurance Equation is defined as:

$$E = \frac{\eta_p}{c_p}\sqrt{\frac{2\rho S}{g}} \cdot \left(\frac{C_L^{\frac{3}{2}}}{C_D}\right) \cdot \left(\frac{1}{\sqrt{m_2}} - \frac{1}{\sqrt{m_1}}\right)$$

Before we can plug into this equation we must first find C_L as a maximum for SLUF using the following relation:

$$C_L = \sqrt{\frac{3C_{D0}}{k}}$$

Using our calculated drag polar parameters:

- $k = 0.0323$
- $C_{D0} = 0.032$

We find that our coefficient of lift is 1.724. Now we can use the remaining parameters to calculate the endurance of our aircraft:

- $C_D = 0.128$
- $\eta_p = 0.8$
- $c_p = 1.2 \times 10^{-6}\ s^{-1}$
- $\rho = 0.653 \frac{kg}{m^3}$
- $g = 9.81 \frac{m}{s^2}$
- $S = \ 25.4\ m^2$
- $W_i = 2658\ kg$
- $W_f = 2242.7\ kg$

Note that for the simplicity of calculation the units have been converted from imperial to metric

Plugging in these values, we obtain an endurance of 10 hours. Comparing this value to existing aircraft, this is a very solid endurance for an STOL aircraft. Similarly, the DHC-3 Otter has an endurance of 9.5 hours.

Power Required

Here we will analyze the performance of our aircraft with the Pratt & Whitney PT6-A 140 turboprop engine. Given the parameters of this engine in the previous sections, we will calculate how much power is required of the engine and how much power is available. We can then use this information to plot the power as a function of velocity and define our maximum velocity. We can use the following relation to define the power available at a certain altitude:

$$P_{available} = \eta_p \cdot P_{Sea\ Level} \cdot \left(\frac{\rho}{\rho_0}\right)$$

This will give us how much power we have available to us to use at a given altitude. We then want to plot the power required for SLUF conditions and find the intersection of the two. We can calculate the power required for SLUF with the following relation:

$$P_{required} = \frac{1}{2}\rho V^3 S C_{D0} + \frac{2kW^2}{\rho V S}$$

Using a MATLAB (Appendix 4.4) script to calculate the power required at 0, 2000, 4000, and 6000 meters we obtain the following plot:

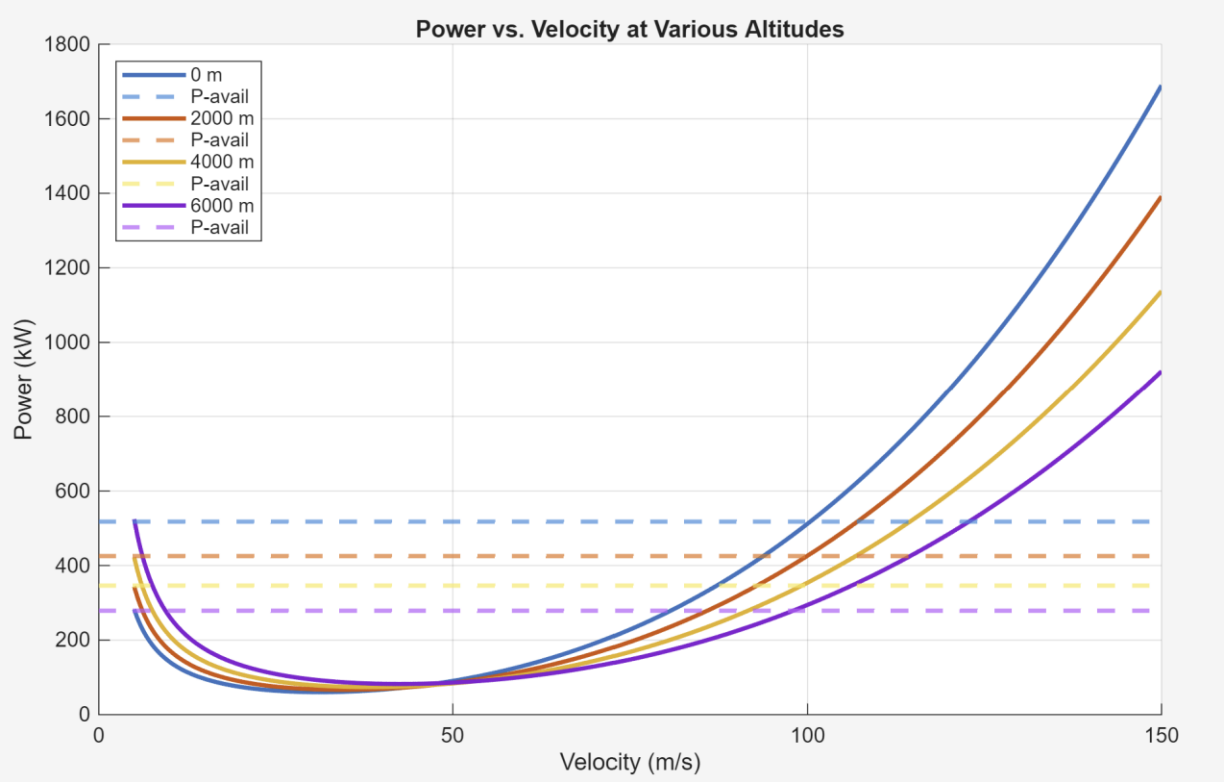


Figure 29. Power vs. Velocity

We also obtain the following results from the script:

```
    Altitude: 0.00 meters
Minimum Velocity: 32.37 m/s
Maximum Velocity: 100.36 m/s

  Altitude: 2000.00 meters
Minimum Velocity: 35.71 m/s
Maximum Velocity: 99.92 m/s

  Altitude: 4000.00 meters
Minimum Velocity: 39.58 m/s
Maximum Velocity: 99.20 m/s

  Altitude: 6000.00 meters
Minimum Velocity: 44.10 m/s
Maximum Velocity: 97.89 m/s
```

From this plot and the results of the script, we can see that the power required and power available falls as altitude increases. We see that near our defined cruising altitude of 20,000 feet (6000 meters) that our maximum velocity is 97.9 m/s or Mach 0.3. This value is

consistent with the intended design of our aircraft in which the goal was to be able to achieve a speed around Mach 0.3. This result also indicates the satisfactory performance of our aircraft with our chosen engine. The combination of this engine with our aircraft design is a strong choice.

Absolute & Service Ceiling

To find the service ceiling of our aircraft, we will use the equation given by [*Anderson, John*] for a good estimate of the service ceiling. This is given by:

$$H = -19{,}867 \cdot ln\left(\frac{W}{(P_{A0})_{\max}}\right) \cdot \sqrt{\frac{2}{\rho_0}\left(\frac{W}{S}\right)} \cdot \left[\frac{0.7436 C_{D0}^{\frac{1}{4}}}{(eAR)^{\frac{3}{4}}}\right]$$

This is the general equation given that English units are used in the computation of this equation. We can plug the following parameters into the above equation to solve for our aircrafts service ceiling:

- $W = 6{,}835\ lbf$
- $P_{A0} = 381{,}480\ lbf \cdot \frac{ft}{s}$
- $\rho_0 = 0.002377$
- $\frac{W}{S} = 36.3\ \frac{lbf}{ft^2}$
- $C_{D0} = 0.032$
- $e = 0.8$ (Oswald Efficiency Factor)
- $AR = 12.3$

Plugging in these parameters to the service ceiling equation, we obtain an absolute service ceiling of 34,373 feet. Although the absolute service ceiling calculation may seem high for an STOL aircraft, we can reconcile this given the use of a more powerful than normal engine than compared to something such as the Cessna Caravan which has almost 200 less shaft-horsepower and weighs 1,500 pounds more.

To estimate the service ceiling of our aircraft, we can use a quick linear approximation to obtain the service ceiling. Using our calculated absolute ceiling, we obtain a service ceiling of 33,308 feet.

Maximum Rate of Climb

To ensure that our engine selection meets the criteria we have been given, the engine must produce enough power to clear 50 feet in altitude in less than 800 feet of distance. We must also be able to do this while landing. To determine our maximum rate of climb, we can plot the rate of climb of our aircraft as a function of velocity. We can use a MATLAB (Appendix 4.4) script at the same time to verify that our aircraft will meet the minimum clearance in less than 800 feet.

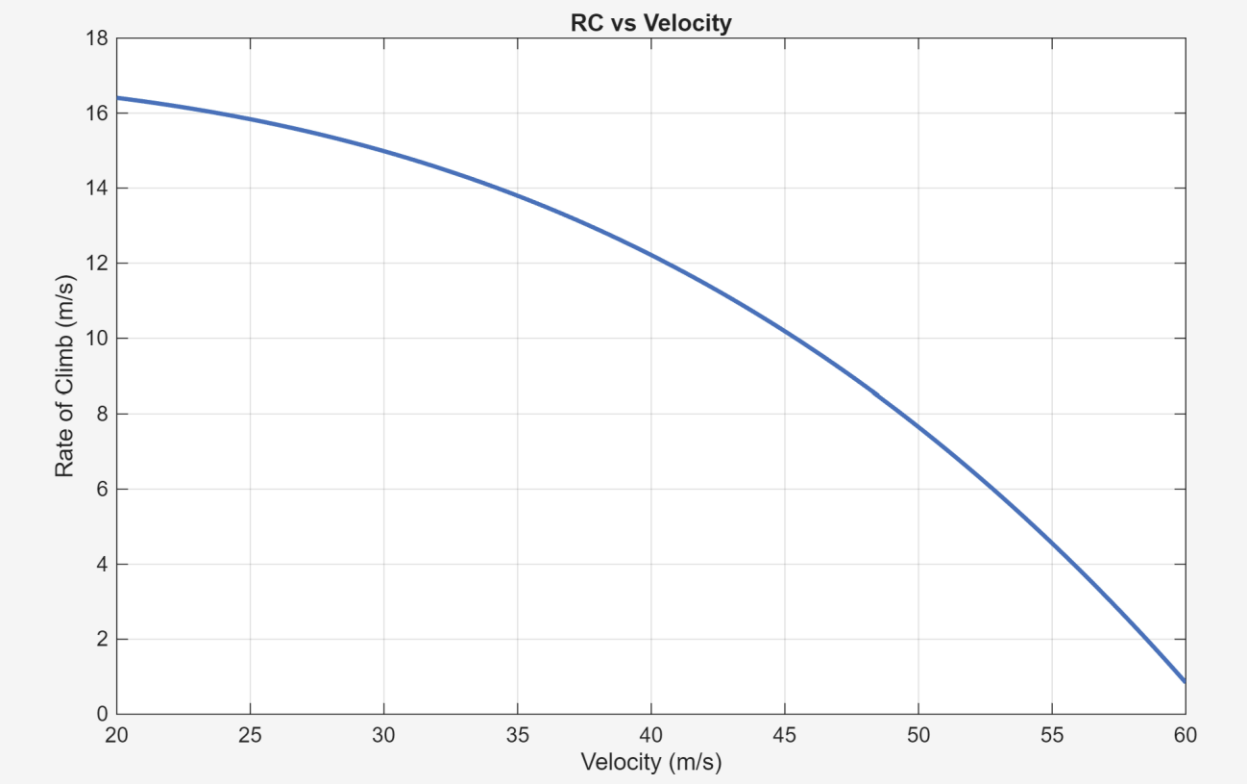


Figure 30. Rate of Climb vs. Velocity

We also obtain the following results from the script:

```
RC = 12.90 m/s (2539 ft/min)
     Gradient = 33.95%
```

```
STOL: PASSED

Max RC = 16.41 m/s at V = 20.0 m/s
```

These results indicate that our aircraft will in fact clear about a 50-foot obstacle in less than 800 feet. These results also show how our aircraft rate of climb will vary with the velocity of the aircraft. The climb performance can be attributed to the engine we have selected for our aircraft. From the graph and the results of the script, we find that the maximum rate of climbing of our aircraft is 16.4 m/s.

Maximum Flight Path Angle

With the use of a NACA 4412 airfoil, we can extract data from external sources to obtain the stall angle of our chosen airfoil. Using *Airfoil Tools,* we can obtain the $C_L \; vs. \; C_D$ and $C_L \; vs. \; \alpha$ plot for our airfoil. We note that for the below plots, a Reynold's number of 10^6 was used to model realistic flight conditions.

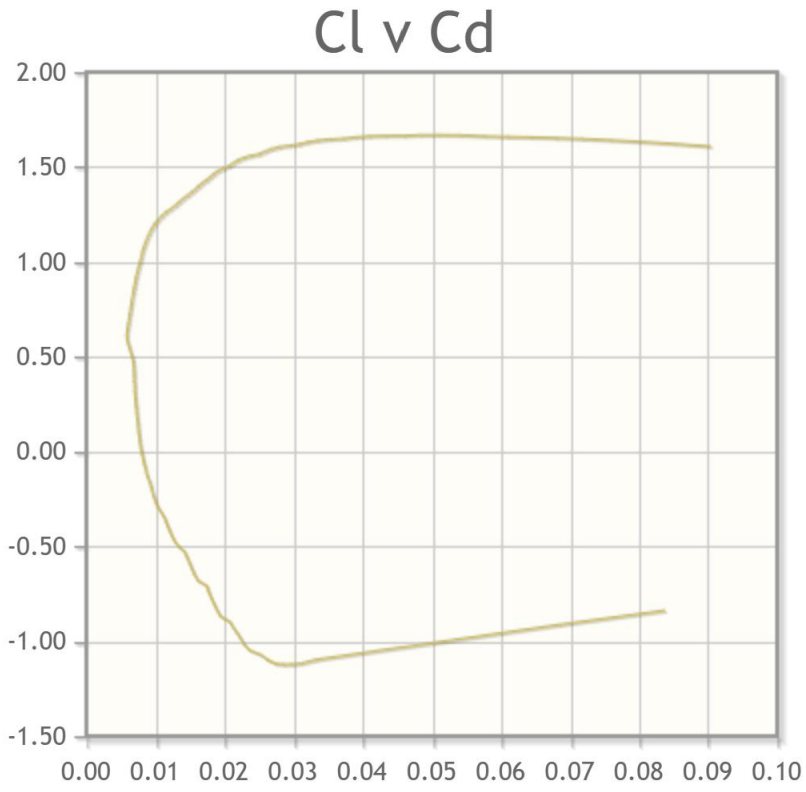


Figure 31. C_L vs C_D Plot for NACA 4412 Airfoil [Airfoil Tools]

In the above plot, we can see that given a positive angle of attack that the maximum coefficient of lift is achieved when the coefficient of drag is approximately 0.05 with a corresponding coefficient of lift of 1.65.

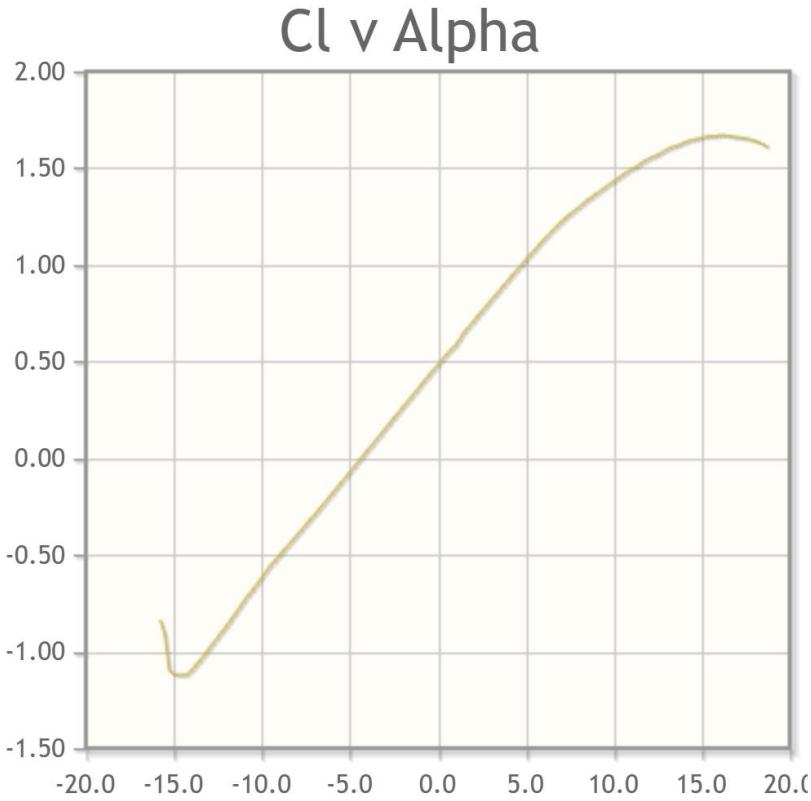


Figure 32. C_L vs α Plot for NACA 4412 Airfoil [Airfoil Tools]

Using the graph in figure 32, we can identify the stall angle to be approximately 16°. Therefore, this will be our aircraft's maximum flight angle. In regular operating conditions, we would restrict the maximum angle of attack to 13 – 14 degrees to allow for a margin of safety and time for the pilots/computer to correct any unsafe action.

5 CONCLUSIONS

In this section, we will discuss and evaluate our chosen design and indicate whether we believe that it will work and perform the given tasks. We will reflect on our process and the lessons that we have learned throughout the semester.

5.1 DESIGN EVALUATION

Requirements Met

After computing and analysis of the performance results of our aircraft, we do believe that our aircraft design concept is going to work. Based on existing short takeoff and landing aircraft, our performance specifications match very reasonably to proven designs. Our aircraft design can reach up to 850 nautical miles under continuous flights, which is comparable to aircraft like the DHC-6 Twin Otter. Our aircraft also features a 10-hour endurance, while optimistic; this endurance is comparable to the Pilatus PC-12.

We wanted to approach our aircraft design with a more simplistic concept to ensure that our aircraft would be up to the given tasks. Many of the systems and design decisions that we have made are tried and tested features that we know will work. To ensure mission timing, we have chosen an engine that we know will produce more than enough power to get the aircraft airborne and get the necessary cargo to its destination in a timely manner.

Overall, our chosen design meets are given requirements and can be summarized as follows:

Table 4. Numerical Requirements Compared to Computed Results

Requirement	Required	Computed
Payload Capacity	2,500 lbs.	2,500 lbs.
Range	400 nm.	850 nm.
Endurance	8 hours	10 hours

Rate of Climb	~2.5 m/s	16.41 m/s
Service Ceiling	~22,000 feet	~34,000 feet

From table 4, we can see that computed performance values are significant enough to meet and exceed the required performance values. In the concept of refinement section (4.3), we also discussed the measures we have taken to mitigate the effects of the harsh Arctic elements. Using existing systems, we believe that we could properly deice the aircraft to keep it in good flying order. The anti-ice systems are mission critical, and the aircraft cannot be deemed airworthy without them.

Our aircraft will be able to fulfill the cargo transport mission and successfully carry the weight equivalent of 9 passengers, an 8,000-watt generator, and 55 gallons of diesel fuel. We have assured ourselves that our aircraft will be able to carry this weight. While performing sizing calculations and overestimating with a payload capacity of 2,500 pounds, we find that we have more than enough power from the engine to fly this mission.

Likewise, we can confirm that we will also be able to fly passenger transport and medical transport by using the same method with our sizing calculations. With verification that we can successfully haul the weight necessary to complete these missions, we are confident that our design will perform strongly.

Remaining Risks

After our initial identification and mitigation of risks, there are some that will still need to be addressed. Although our aircraft should be able to land in the Arctic conditions, runway ice could still play a key factor. The runway wherever the mission may take the aircraft must be deemed within safe limits to attempt a landing. Too much ice could pose a serious safety risk to those on the ground and those in the aircraft.

Another risk that needs to be addressed also concerns the cold operating conditions. The battery and electrical system will be prone to failure given low temperatures. These systems

will need to be warmed, especially while the aircraft is on the ground. The battery system will be responsible for ensuring that the aircraft systems remain fully operational while the engine is off. This system will also be needed as an emergency backup in case of engine failure.

5.2 NEXT STEPS

If we were to continue into the next semester, the next thing that we would research is the costs associated with our design. Our design could be flawless for the mission, but if it is way overbudget then what is the point of having a flawless design? We would need to keep researching and see if the design choices that we made were within a realistic cost to produce this aircraft. If not, then the next step would be to go back and rework whatever needs to be to get the aircraft to also be cost efficient.

We could also then start looking into manufacturing techniques and how we can efficiently and cost-effectively produce our aircraft. It would be smart to investigate different companies that produce aircraft components to see if it would be smart to use domestic suppliers or international suppliers. Depending on how fast the stakeholders need this aircraft to be operational, it would also be smart to consider how fast different techniques or companies could get us the supplies/components.

A future step for this aircraft design would be to consider the several types of instruments and avionics that could be placed in the cockpit. Considerations such as steam gauges versus a fully digital display or mechanical linkages versus fly-by-wire. These considerations would also play into the cost considerations as a fully digital display could be significantly more expensive than steam gauges. However, a system such as a digital display would offer the pilot more tools and resources to safely navigate the aircraft.

Structural design, while beyond the scope of this project, would also have to be another consideration. With the right structural design, we can save weight on the fuselage and increase the wight of the payload that we can carry. Structural rigidity is also especially important to reinforce the landing gear and to decrease the risk to the passengers/cargo in an emergency landing. Choice of material is also extremely important as it plays into

structural strength, weight, safety, and performance. For example, if we were to choose a material with a higher modulus of elasticity, the fuselage would be more resistant to deformation and aerodynamic loads.

Following some of the more important design considerations, we would move on to considering the interior of the aircraft. For all mission types, a modular floor would make aircraft turnarounds much faster between several types of missions. The floor would assist with securing cargo and medical or passenger transport; the necessary equipment and seats could be mounted to this floor. Other considerations such as an oxygen system would also be important for passenger safety.

5.3 LESSONS LEARNT

Throughout this project we have learned many things about larger scale projects such as this one. The design process for aircraft is a challenging one and for good reason, without meticulous attention to detail and ensuring that every factor is accounted for there probably would be no such thing as an aircraft. Being able to see the larger picture and how every decision and formula fits in has been the biggest thing learned throughout this project.

Time management has been another aspect of this project that we have had to refine and manage throughout this project. Balancing the project milestones with the assignments where we learned the theory was crucial to an effective design. We would communicate effectively as a team to ensure that all aspects of our design were being covered and that while working independently, all our data was matching up and supporting the design concept that we were working toward.

We also learned that the aircraft is highly interconnected, with calculations such as weight, fuel fraction, and aerodynamic performance all depending on one another. A change to a single parameter can significantly impact other aspects of the design. For example, we were struggling with the payload weight, influencing the range of aircraft to allow all missions to be operational. This problem, like many others we encountered during the design process, makes that process iterative and requires careful optimization to balance completing requirements.

Another key lesson comes from just how mission requirements shape every aspect of the aircraft. Small assumption changes such as range or payload had a major impact on aspects such as the takeoff weight, fuel fraction, and wing size. Another important takeaway was the need to balance competing design requirements, particularly between STOL performance and cruise efficiency, which required careful selection of wing loading, aspect ratio, and airfoil characteristics. Having practical assumptions and reference aircraft really helped us through our design process and led us to have realistic results.

There are many variables to adjust when designing an aircraft. Determining criteria and constraints of the aircraft helped to narrow down the final design, however still left much to be decided. We looked at previous models in the past to help give a path on where to start designing this aircraft. Through our design process, we constantly iterated throughout our learning of more improved content and more precision for different calculations. The iterative design process helped us design the best aircraft that would fit the mission profile.

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APPENDIX: MATLAB CODE

SIZING CODE (IT 1 + 2)

Original code:

```
%AAE 251 Spring 2026
%MP4
%Aircraft Sizing
%Authors:  Team A203
%GTOW sizing script for the 3 aircraft
eta_p = 0.85;      % prop efficiency
LD    = 13;        % representative L/D for small STOL/utility aircraft
c_hr  = 0.50;      % TSFC (1/hr) representative turboprop sizing
E_hr  = 0.50;      % loiter/reserve time (hr)
trap  = 1.06;      % trapped fuel factor
% Segment fractions
W_warmup  = 0.995;
W_takeoff = 0.995;
W_climb   = 0.980;
W_desc    = 0.990;
W_land    = 0.992;
% Empty-weight fraction model We/W0 = A*W0^C
A = 2.05;
C = -0.18;
% Crew weight assumption
W_crew = 400; % lb (2 crew)
% Aircraft 1: DHC-3 Otter
% From PM4: payload 2,100 lb, cruise 120 kt. (Range ~820 nm typical)
W_payload = 2100;              % lb  (PM4)
V_kts     = 120;               % kt  (PM4)
R_nm      = 820;               % nm  (typical Otter max range)
W0_guess  = 8000;              % lb  (reference MTOW)
```

```
[W0_otter, Wf_otter] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
                        W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                        A,C,W_payload,W_crew,W0_guess);
fprintf('DHC-3 Otter: W0 ~= %.0f lb | Fuel frac = %.4f | Ref MTOW ~ 8000 lb\n\n', ...
    W0_otter, Wf_otter);
% Aircraft 2: Cessna 208B Grand Caravan EX
% From PM4: max payload 3,600 lb, range 494 nm, max cruise ~194 kt (BJ).
W_payload = 3600;             % lb  (PM4)
V_kts     = 194;              % kt  (PM4/BJ)
R_nm      = 494;              % nm  (PM4/BJ)
W0_guess  = 8807;             % lb  (reference MTOW)
[W0_caravan, Wf_caravan] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
                        W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                        A,C,W_payload,W_crew,W0_guess);
fprintf('Cessna 208B Caravan EX: W0 ~= %.0f lb | Fuel frac = %.4f | Ref MTOW = 8807
lb\n\n', ...
    W0_caravan, Wf_caravan);
% Aircraft 3: Pilatus PC-6 Porter
% From PM4: useful load ~ 2,500 lb.
% Typical: cruise ~125 kt, range ~500 nm, MTOW ~6173 lb (PC-6/B2-H4).
W_payload = 2500;             % lb  (PM4)
V_kts     = 125;              % kt  (typical)
R_nm      = 500;              % nm  (typical)
W0_guess  = 6173;             % lb  (reference MTOW)
[W0_pc6, Wf_pc6] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
                        W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                        A,C,W_payload,W_crew,W0_guess);
fprintf('Pilatus PC-6 Porter: W0 ~= %.0f lb | Fuel frac = %.4f | Ref MTOW ~ 6173 lb\n\n', ...
    W0_pc6, Wf_pc6);
% Function
function [W0, Wf_frac] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
```

```
                    W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                    A,C,W_payload,W_crew,W0_guess)
  % Unit conversions
  R_ft   = R_nm * 6076;            % nm -> ft
  V_ft_s = V_kts * 1.68781;        % kt -> ft/s
  c_s    = c_hr / 3600;            % 1/hr -> 1/s
  % Cruise weight fraction (prop Breguet exponential)
  W_cruise = exp( - (R_ft * c_s) / (eta_p * LD * V_ft_s) );
  % Loiter weight fraction
  W_loiter = exp( - (E_hr * c_hr) / (eta_p * LD) );
  % Total mission fraction
  M_ff = W_warmup * W_takeoff * W_climb * W_cruise * W_loiter * W_desc * W_land;
  % Fuel fraction (with trapped fuel)
  Wf_frac = trap * (1 - M_ff);
  % Iterate GTOW
  W0 = W0_guess;
  for i = 1:200
     We_W0 = A * (W0^C);
     W0_new = (W_payload + W_crew) / (1 - Wf_frac - We_W0);
     if abs(W0_new - W0) < 1
        W0 = W0_new;
        return;
     end
     W0 = W0_new;
  end
End
```

Iteration 2 (more parameters):

```
% AAE 251 Spring 2026
```

```
% MP4
% Aircraft Sizing
% Authors: Team A203
% GTOW sizing script for the aircraft
eta_p = 0.85;      % prop efficiency
LD    = 20;        % representative L/D for NACA 4412
% c_hr  = 0.45;      % TSFC (1/hr)
E_hr  = 0.50;      % loiter/reserve time (hr)
trap  = 1.06;      % trapped fuel factor
rho_fuel = 6.7;    % Fuel Density lb/gal
% Segment fractions
W_warmup  = 0.995;
W_takeoff = 0.995;
W_climb   = 0.985;  % slightly improved for STOL aircraft
W_desc    = 0.990;
W_land    = 0.992;
% Empty-weight fraction model We/W0 = A*W0^C
A = 2.05;
C = -0.18;
% Crew weight assumption
W_crew = 400; % lb (2 crew)
%% Aircraft 1: DHC-3 Otter
W_payload = 2600;
V_kts     = 120;
R_nm      = 400;
W0_guess  = 8000;
[W0_otter, Wf_otter, Vfuel_otter] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
                    W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                    A,C,W_payload,W_crew,W0_guess,rho_fuel);
fprintf('NACA 4412: W0 ~= %.0f lb | Fuel frac = %.4f | Fuel Vol = %.0f gal\n\n', ...
    W0_otter, Wf_otter, Vfuel_otter);
```

```
%% Function
function [W0, Wf_frac, V_fuel_gal] = size_W0_prop(R_nm,V_kts,E_hr,c_hr,eta_p,LD,trap, ...
                        W_warmup,W_takeoff,W_climb,W_desc,W_land, ...
                        A,C,W_payload,W_crew,W0_guess,rho_fuel)
 % Unit conversions
 R_ft   = R_nm * 6076;
 V_ft_s = V_kts * 1.68781;
 c_s    = c_hr / 3600;
 % Cruise weight fraction
 W_cruise = exp( - (R_ft * c_s) / (eta_p * LD * V_ft_s) );
 % Loiter weight fraction
 W_loiter = exp( - (E_hr * c_hr) / (eta_p * LD) );
 % Total mission fraction
 M_ff = W_warmup * W_takeoff * W_climb * W_cruise * W_loiter * W_desc * W_land;
 % Fuel fraction
 Wf_frac = trap * (1 - M_ff);
 % Iterate GTOW
 W0 = W0_guess;
 for i = 1:200
    We_W0 = A * (W0^C);
    W0_new = (W_payload + W_crew) / (1 - Wf_frac - We_W0);
    if abs(W0_new - W0) < 1
       W0 = W0_new;
       break;
    end
    W0 = W0_new;
 end
 % Fuel weight and volume
 W_fuel = Wf_frac * W0;
 V_fuel_gal = W_fuel / rho_fuel;
end
```

Determination of $C_{D,0}$ (3.6)

```
% AAE 251 - Initial CD0 Estimate

%% Reference area

S_ref = 273.4;   % ft^2  (wing reference area)

%% Skin friction coefficient guess

% Typical conceptual-design turbulent average value

Cf = 0.0030;

%% --------------------------------

% Wetted areas (ft^2)

%% --------------------------------

S_wet_wing = 2.0 * S_ref;     % common first estimate

S_wet_fuse = 750;             % estimated fuselage wetted area

S_wet_ht   = 2.0 * 71;        % if Sh = 71 ft^2

S_wet_vt   = 120;             % rough first estimate

S_wet_nac  = 90;              % two engine nacelles combined

%% --------------------------------

% Form factors

%% --------------------------------

FF_wing = 1.25;

FF_fuse = 1.10;
```

```
FF_ht  = 1.20;

FF_vt  = 1.20;

FF_nac = 1.30;

%% --------------------------------

% Interference factors

%% --------------------------------

Q_wing = 1.00;

Q_fuse = 1.00;

Q_ht  = 1.05;

Q_vt  = 1.05;

Q_nac = 1.20;

%% --------------------------------

% Component drag-area contributions

%% --------------------------------

Dq_wing = Cf * FF_wing * Q_wing * S_wet_wing;

Dq_fuse = Cf * FF_fuse * Q_fuse * S_wet_fuse;

Dq_ht  = Cf * FF_ht  * Q_ht  * S_wet_ht;

Dq_vt  = Cf * FF_vt  * Q_vt  * S_wet_vt;

Dq_nac = Cf * FF_nac * Q_nac * S_wet_nac;

%% Landing gear / misc parasite drag
```

```
% For fixed gear, this is often added separately as a drag area

Dq_gear_misc = 1.8;   % ft^2, reasonable first-pass guess for fixed gear + misc items

%% Total parasite drag area

Dq_total = Dq_wing + Dq_fuse + Dq_ht + Dq_vt + Dq_nac + Dq_gear_misc;

%% CD0

CD0 = Dq_total / S_ref;

%% Outputs

fprintf('Wing contribution drag area      = %.3f ft^2\n', Dq_wing);

fprintf('Fuselage contribution drag area  = %.3f ft^2\n', Dq_fuse);

fprintf('H-tail contribution drag area    = %.3f ft^2\n', Dq_ht);

fprintf('V-tail contribution drag area    = %.3f ft^2\n', Dq_vt);

fprintf('Nacelle contribution drag area   = %.3f ft^2\n', Dq_nac);

fprintf('Gear/misc drag area              = %.3f ft^2\n', Dq_gear_misc);

fprintf('Total drag area                  = %.3f ft^2\n', Dq_total);

fprintf('Estimated CD0                    = %.4f\n', CD0);
```

Coefficient of Drag vs. Flight Speed Code (4.4)

```
%Constants
air_density = 0.001267;     %[slugs/ft^3]  ~Air density at 20,000 feet
weight = 6835;              %[lbs] Weight of the aircraft
wing_area = 273.4;          %[ft^2] ~Wing surface area
```

```
C_D0 = 0.032;            %[] Zero lift drag coefficient
k = 0.0323;              %[] Induced drag coefficient

%Velocity Vector
v = linspace(100,600,1000);  %[ft/s]

%Calculate CD as a function of flight speed
C_D = C_D0 + (4 * k * weight^2) ./ (air_density^2 * wing_area^2 .* v.^4);

%Plot CD as a function of flight speed
figure(1)
hold on
title("C_D vs. Flight Speed")
xlabel("Flight Speed (ft/s)")
ylabel("C_D")
plot(v, C_D, lineWidth=2)
grid on
hold off
```

Power vs. Velocity (4.4)

```
%PM9 Thrust and Power Modeling

%Constants
S = 25.4;       %[m^2] Wing Surface Area
CD0 = 0.032;    %[] Zero Lift Drag Coefficient
k = 0.0323;     %[] Drag Polar Constant
e = 0.8;        %[] Oswald Efficiency Factor
AR = 15;        %[] Aspect Ratio
rho_0 = 1.225;  %[kg/m^3] Sea-Level Air Density
m_MTOW = 2658;  %[kg] Maximum Takeoff Weight
g = 9.81;       %[m/s^2] Acceleration Due to Gravity

%Pratt & Whitney PT6A-140 Engine Parameters
P_SL = 647000;  %[W] Sea-Level Takeoff Power
```

```
eta = 0.8;      %[] Propeller Efficiency

%Define Measureable Altitudes
alts = [0 2000 4000 6000];  %[m] Various Altitudes

%Define Velocity Vector
V = linspace(5, 150, 1000);   %[m/s] Flight Velocity

figure(1)
hold on
title("Power vs. Velocity at Various Altitudes")
xlabel("Velocity (m/s)")
ylabel("Power (kW)")

colors = lines(length(alts));

for i = 1:length(alts)

   %Calculate Air Density at Given Altitude
   rho = rho_0 * ((1 - (alts(i) / 44330)) ^ 4.256);

   %Calculate Power Availible
   P_available = eta * P_SL * (rho / rho_0);

   P_required = zeros(size(V));

   %Calculate Power Required for SLUF
   for j = 1:length(V)

      P_required(j) = (0.5 * rho * V(j)^3 * S * CD0) + ((2 * k * (m_MTOW * g)^2) / (rho * V(j) *
S));

   end

   [~, idx] = min(abs(P_required - P_available));
   V_max = V(idx);

   CLmax = 1.6;  %For NACA 4412 Airfoil
   V_min = sqrt((2 * g * m_MTOW) / (rho * S * CLmax));

   plot(V, (P_required / 1000), LineWidth = 2)
   yline((P_available / 1000), "--", "Color", colors(i,:),LineWidth = 2)

   fprintf("Altitude: %0.2f meters\n", alts(i));
   fprintf("Minimum Velocity: %0.2f m/s\n", V_min);
   fprintf("Maximum Velocity: %0.2f m/s\n\n", V_max)
```

```
end

legend('0 m', 'P-avail', '2000 m', 'P-avail', '4000 m', 'P-avail', '6000 m', 'P-avail', 'Location',
'northwest');
grid on
hold off
```

Drag Polar Plot (4.4)

```
% AAE 251 - Drag Polar
% Inputs
CD0 = 0.032;       % zero-lift drag coefficient
b   = 58;          % ft, estimated wingspan
S   = 273.4;       % ft^2, wing area
e   = 0.80;        % Oswald efficiency factor
% Aspect ratio
AR = b^2 / S;
% Induced drag factor
k = 1 / (pi * e * AR);
% CL range
CL = linspace(0, 2.5, 300);
% Drag polar
CD = CD0 + k * CL.^2;
% Plot
figure
plot(CL, CD, 'LineWidth', 2)
grid on
xlabel('C_L')
ylabel('C_D')
title('Aircraft Drag Polar')
```

Determination of Drag Polar (3.6)

```
% AAE 251 - Drag Polar
```

```
% Inputs
CD0 = 0.032;       % zero-lift drag coefficient
b   = 58;          % ft, estimated wingspan
S   = 273.4;       % ft^2, wing area
e   = 0.80;        % Oswald efficiency factor
% Aspect ratio
AR = b^2 / S;
% Induced drag factor
k = 1 / (pi * e * AR);
% CL range
CL = linspace(0, 2.5, 300);
% Drag polar
CD = CD0 + k * CL.^2;
% Plot
figure
plot(CL, CD, 'LineWidth', 2)
grid on
xlabel('C_L')
ylabel('C_D')
title('Aircraft Drag Polar')
```

```
% Display values

fprintf('Aspect Ratio = %.2f\n', AR);

fprintf('Oswald Efficiency Factor = %.2f\n', e);

fprintf('k = %.4f\n', k);

fprintf('Drag Polar: CD = %.3f + %.4f CL^2\n', CD0, k);
```

Maximum Rate of Climb (4.4)

```
%% Climb Performance Calculation (UPDATED)
%% --- INPUTS ---
P_engine_hp = 867;   % PT6A-140
eta_p = 0.8;
W0_kg = 3100.304;
rho = 1.4;   % Arctic air
V = 38; %initial guess climb speed
S = 25.4;
CD = 0.128; % your value
g = 9.81;
%% --- CONVERSIONS ---
P_engine = P_engine_hp * 745.7;
W = W0_kg * g;
%% --- POWER AVAILABLE ---
P_avail = eta_p * P_engine;
%% --- DRAG ---
D = 0.5 * rho * V^2 * S * CD;
%% --- POWER REQUIRED ---
P_req = D * V;
%% --- RATE OF CLIMB ---
RC = (P_avail - P_req) / W;
```

```
RC_ftmin = RC * 196.85;
%% --- CLIMB GRADIENT ---
gradient = RC / V;
gradient_percent = gradient * 100;
%% --- STOL CHECK ---
required_gradient = 50 / 800;
meets_STOL = gradient > required_gradient;
fprintf('RC = %.2f m/s (%.0f ft/min)\n', RC, RC_ftmin);
fprintf('Gradient = %.2f%%\n', gradient_percent);
if meets_STOL
  fprintf('STOL: PASSED\n');
else
  fprintf('STOL: FAILED\n');
end
%% --- FIND BEST CLIMB SPEED
V_range = linspace(20, 60, 200);
RC_range = zeros(size(V_range));
for i = 1:length(V_range)
  D_i = 0.5 * rho * V_range(i)^2 * S * CD;
  P_req_i = D_i * V_range(i);
  RC_range(i) = (P_avail - P_req_i) / W;
end
[RC_max, idx] = max(RC_range);
V_best = V_range(idx);
fprintf('\nMax RC = %.2f m/s at V = %.1f m/s\n', RC_max, V_best);
figure;
plot(V_range, RC_range, 'LineWidth', 2);
xlabel('Velocity (m/s)');
ylabel('Rate of Climb (m/s)');
title('RC vs Velocity');
grid on;
```